Internet Appendix for "The Intramonth Momentum Cycle"

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This Internet Appendix contains supplementary material for "The Intramonth Momentum Cycle." It includes (IA.1) Data Construction and Sample Formation, (IA.2) Benchmarked Contribution and Recovery Estimands, (IA.3) T+1 Settlement Falsification Tests, (IA.4) Tail Trimming Analysis, (IA.5) Crash-Day Concentration Methodology, (IA.6) International Evidence: Methodology, (IA.7) Historical PreTOM Concentration, (IA.8) Mutual Fund Cash Buffers, (IA.9) Recovery of Selling-Associated Losses: Specifications, (IA.10) Recovery Response under Equal Weighting, (IA.11) Long-Horizon Path of New Losers, (IA.12) Same-Weekday Momentum and the Window, (IA.13) Is the Window Selling Informed?, (IA.14) Window Dressing Falsification, (IA.15) 52-Week-High Sort: Additional Evidence, (IA.16) Beta Controls, (IA.17) European T+3-to-T+2 Settlement Transition, (IA.18) U.S. T+3-to-T+2 Settlement Transition, (IA.19) S&P 500 Membership Robustness, (IA.20) The Window, (IA.21) Supporting Tables and Figures for the Main Text, (IA.22) Alternative Benchmarks, (IA.23) Acute Liquidity Shocks, and (IA.24) The Momentum Factor and Fund Persistence.

IA.1 Data Construction and Sample Formation

The daily stock-level panel used in the main analysis is constructed as follows.

1. **CRSP input data.** We obtain the CRSP daily stock file from WRDS in Flat File For-

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mat 2.0 (CIZ). The file spans December 29, 1978 through December 31, 2025.[1] Every estimate in the paper that begins in 1980 is built from this extract. The historical comparisons that reach before 1980, which are the era rates and the subperiod tables of Section IA.7, come from a second build, `crsp_1927-2025_fixed_sorting_full.parquet`, which applies the same fixed monthly sorting rule to the full CRSP daily history and holds 73.6 million stock-days from January 3, 1927 to December 31, 2025. Both builds apply the same fixed monthly sorting rule. They are separate files, and the pre-1980 rows exist only in the second. All WRDS inputs were downloaded in March 2026; WRDS data are revised periodically, so later downloads may yield slightly different raw data.

2. **CRSP sample restrictions.** We restrict the CRSP sample to U.S. common operating-company equities listed on the NYSE, AMEX, or Nasdaq, and retain only regular-way, active observations.[2]

3. **Trading-day panel construction.** Using the CRSP trading calendar, we expand each security's panel so that all trading days between its first and last observed trading day are included. Trading days without an observed CRSP record remain in the panel as missing rows.

4. **Month-relative trading-day index.** Each trading day is indexed relative to the turn-of-the-month (TOM). Within each calendar month, the last trading day is labeled τ, the second-to-last $\tau-1$, and so on through $\tau-9$ for the tenth-to-last trading day. Trading days at the start of a calendar month are indexed forward from the preceding month's TOM: $\tau+1$ is the first trading day of the calendar month, $\tau+2$ the second, and so on. The early-month controls $[\tau+5, \tau+8]$ used in the main paper are therefore the fifth through eighth trading days of the calendar month, counted from the preceding TOM.

[1] Observations from 1978–1979 are retained in order to construct momentum signals for the first valid holding months in 1980.

[2] In the CIZ format, these restrictions are implemented sequentially as follows: `ShareType='NS'`, `SecurityType='EQTY'`, `SecuritySubType='COM'`, `USIncFlg='Y'`, `IssuerType` in {`ACOR`, `CORP`}, `PrimaryExch` in {`N`, `A`, `Q`}, `ConditionalType='RW'`, and `TradingStatusFlg='A'`. Legacy CRSP share-code filters such as 10 and 11 are not available in the same way in CIZ, so we use the equivalent CIZ security-type and trading-status fields instead.

5. **Momentum construction.** For each stock-month, we compute 12–2 momentum from monthly returns aggregated from daily CRSP observations:

$$M_{i,m} = \prod_{s=m-12}^{m-2} (1 + r_{i,s}) - 1.$$

The resulting stock-month signal is assigned to all daily observations in holding month m.

6. **Momentum validity.** A momentum window is treated as valid only if it contains no trading-calendar-added empty rows and no missing-return indications according to the CRSP variable `DlyRetMissFlg`.[3] We then retain only observations with valid momentum and non-missing lagged market capitalization. The final analysis panel used in the paper contains 53,324,145 observations on 22,744 securities, spanning January 2, 1980 through December 31, 2025.

7. **Momentum deciles and portfolio weights.** We assign each stock-month to a momentum decile using Kenneth French's NYSE 2–12 prior-return breakpoints, aligned to the month immediately preceding the holding month. All stock-months that survive the post-momentum filters are successfully matched to a breakpoint date and receive a non-missing decile assignment. For the portfolio analysis, weights are computed within each (*date*, *decile*) cell using lagged market capitalization:

$$w_{i,t}^{(q)} = \frac{\mathrm{MCAP}_{i,t-1}}{\sum_{j \in q(t)} \mathrm{MCAP}_{j,t-1}},$$

where $q(t)$ denotes the set of securities in the relevant decile on date t, and $\mathrm{MCAP}_{i,t-1}$ is security i's market capitalization on the previous trading day. These lagged market-cap weights are included in the final analysis panel.

8. **Additional daily characteristics.** We merge S&P 500 membership from the CRSP index constituents database. We also merge daily factors from Kenneth French's data library (*Fama/French 3 Factors*), from which we use the daily risk-free rate (`RF`) and

[3] In the baseline specification, a day is admissible only if `DlyRetMissFlg` is either absent or equal to `NA`. Thus, windows are excluded if they contain any day with `DlyRetMissFlg` in {`MV`, `NS`, `NT`, `RA`, `GP`, `MP`, `DG`, `DM`, `DP`}.

the daily market return (`Mkt`) in the analysis. This is the series the paper calls the value-weighted market throughout. It is built from CRSP but applies French's own universe screens, so it is not identical to a value-weighted index of the stocks in our panel. In addition, we construct the bid-ask spread measure

$$\mathrm{BAS}_{i,t} = \frac{\mathrm{Ask}_{i,t} - \mathrm{Bid}_{i,t}}{(\mathrm{Ask}_{i,t} + \mathrm{Bid}_{i,t})/2}.$$

The S&P 500 membership indicator is available for all observations; 5,460,281 daily rows correspond to index members and 47,863,868 correspond to non-members. The BAS variable is non-missing for 46,560,017 out of 53,324,145 observations, corresponding to 87.32% coverage.

9. **TAQ sell-pressure measures.** We merge daily sell-pressure measures derived from WRDS TAQ intraday indicator data into the CRSP panel using `PERMNO`-date. The TAQ data are available from September 10, 2003 through October 27, 2022, so coverage is nonzero only within this period. The merge is performed as a left merge, so the number of daily panel rows is unchanged. Within the full 1980–2025 panel, 17,861,276 observations have non-missing net selling pressure, corresponding to 33.50% of the sample. Restricting to the TAQ-available period, the panel contains 18,217,051 observations on 9,416 securities, of which 98.05% have non-missing sell-pressure measures.

10. **Final panel.** The final dataset is a daily `PERMNO`-date panel spanning January 2, 1980 through December 31, 2025. It contains CRSP returns and market-cap variables, momentum measures and momentum deciles, the month-relative trading-day index, lagged market-cap portfolio weights, S&P 500 membership, daily factor variables, bid-ask spreads, and TAQ-based sell-pressure measures.

Mutual-fund sample. Holdings are the Thomson Reuters S12 mutual-fund equity holdings file, not the S34 file collected from Form 13F, and are matched to CRSP mutual-fund data through MFLINKS. For the fund-level analysis we restrict to actively managed U.S. equity diversified funds via CRSP's objective code (`crsp_obj_cd LIKE 'ED%'`), excluding ETF share classes (`et_flag = 'F'`) and ETNs (`et_flag = 'N'`), as well as any fund flagged as an index fund under the CRSP categorical scheme (`index_fund_flag` $\in$ {`D, E, B, BMT`}, cov-

ering pure index, enhanced index, index-based, and benchmark-matched products). Fund flow rates are the Coval–Stafford implied flow, $F_{f,m} = TNA_{f,m} - TNA_{f,m-1}(1 + r_{f,m})$, summed over the holdings-reporting interval and divided by fund assets at its start, with total net assets summed and returns asset-weighted across all CRSP share classes of the same fund before the flow is computed. The rate is winsorized at the 1st and 99th percentiles within report date, and fund-months in which the number of covered share classes changes, or in which the implied rate exceeds 100% in absolute value, are treated as unobserved, since total net assets also jump when one fund absorbs another. Share counts are put on a common basis with the CRSP cumulative share adjustment factor. Excess liquidation compares the change in a position with the change that scaling the whole portfolio by the fund's flow would have produced. Let Q_0 and Q_1 be the shares held at the start and end of the reporting interval, put on a common basis with the CRSP cumulative share adjustment factor, and let ϕ be the fund's signed flow over that interval as a fraction of assets at its start, so that $\phi < 0$ under net outflow. Then

$$EL = \phi - \left(\frac{Q_1}{Q_0} - 1 \right),$$

a fraction of the beginning position: the flow-adjusted share-reduction benchmark sets $Q_1/Q_0 - 1 = \phi$ and gives $EL = 0$, while a position closed outright sets $Q_1 = 0$ and gives $EL = 1 + \phi$. That equality defines the benchmark; it is not a general description of proportional selling, since returns and redemptions both occur within the interval. The common flow offset ϕ is absorbed by the fund-report fixed effects, so the coefficients compare share reduction across momentum groups within a report. Coefficients on EL are reported in percentage points, that is as $100\,EL$. At the fund-report level, the outflow magnitude *Outflow* of Table 11 has mean 3.51% and standard deviation 6.08%, with percentiles 1.20%, 4.42%, 9.51%, 15.10% and 30.05% at the 50th, 75th, 90th, 95th and 99th.

Daily fund returns. The daily fund returns of Section IA.24 apply the same universe filter and additionally require assets of at least $10 million, since the daily net-asset-value sample is dominated by very small share classes. The threshold does not affect the estimates.

Extreme daily returns. CRSP's daily file contains a small number of very large returns for distressed microcaps, some of which reflect corporate actions that its own adjustment factors do not capture. In the 1980–2025 sample there are 3,771 daily returns above 100 percent in absolute value, 0.0071 percent of observations, of which 2,678 fall in the loser decile and 721 in the loser decile during the window. Their median price is \$1.50 and their median market capitalization is \$5.6 million against \$142 million for the panel as a whole, so they carry 0.0002 percent of the value weight and full weight when observations are weighted equally. The largest, a return of 3,972 percent, is reproduced from CRSP without modification, with an unchanged price adjustment factor and unchanged shares outstanding. We retain these observations in the reported specifications and report here what happens without them. Winsorizing daily returns at the 0.1 and 99.9 percentiles, or dropping stock-months that contain any daily return above 100 percent in absolute value, leaves the stock-level conclusions in place, and no estimate changes sign. The Loser $\times$ PreTOM coefficient of Table 2 moves from -7.08 to -7.18 and -7.03 basis points per day value-weighted and from -2.51 to -2.33 and -2.41 equal-weighted; the window-specific recovery slope of Table IA.6 moves from -0.0148 to -0.0126 and -0.0110; the within-loser 52-week-high window spread of Table 12 stays negative under both rules, at -8.2 basis points per day in the baseline, -14.2 after winsorization and -9.8 after dropping the extreme stock-months, with corresponding window-minus-rest differentials of -6.2, -6.6 and -5.7 (on the total-return high construction, not the price high of Table 12); the stock-level settlement contrast of Table 8 moves from 196.3 to 196.7 and 196.6, with the same t-statistic to two decimals in all three; and the value-weighted window abnormal return under stock-by-month and date fixed effects moves from 55.4 to 55.1.

IA.2 Benchmarked Contribution and Recovery Estimands

Aggregate calendar returns are consistent with any degree of reversal. A loser portfolio that falls in the window and rises afterward looks the same whether the rise repays the window loss or is a return the portfolio would have earned anyway. The within-cohort comparison distinguishes the two, because recovery of selling-related losses must accrue to the stocks that lost most, while a common rebound does not. It does not: about 7 percent of the cross-sectional window loss associated with differential selling reverses over the turn of the month, and full recovery is rejected (Section 3.1 of the paper). The window's contribution to the premium is therefore reported as a benchmarked quantity, and this section sets out the model that makes the distinction precise.

Index months $m = 1, \ldots, M$ and partition each month into mid-month D, the window W, and Post O, in calendar order $D \to W \to O$. Month-end selling adds a pressure p_m to the leg in month m's window; a fraction θ_O is given back in the Post window that follows it, and a fraction θ_D during the next month's mid-month, so that $\theta = \theta_O + \theta_D$ is the share reversed within the horizon. Mid-month precedes the window inside its own month, so the reversal a mid-month carries is of the previous month's pressure. The window rates are unrestricted:

$$\begin{aligned} R^W_m &= \mu_W + p_m + \varepsilon^W_m, \\ R^O_m &= \mu_O - \theta_O\, p_m + \varepsilon^O_m, \\ R^D_m &= \mu_D - \theta_D\, p_{m-1} + \varepsilon^D_m. \end{aligned} \tag{IA.1}$$

Here μ_W, μ_O and μ_D are whatever the leg earns in each window absent the selling, including any turn-of-the-month cycle, beta exposure, or calendar drift. The object of interest is the window's contribution to the monthly premium, $(1-\theta)\pi$, where $\pi = E[p_m]$.

The benchmarked contribution. The counterfactual return of the same loser portfolio on the same six calendar days absent the selling is unobserved. The three window means give three moments while the model has six parameters, $(\mu_W, \mu_O, \mu_D, \pi, \theta_O, \theta_D)$, so $(1-\theta)\pi$ is not point identified from calendar means alone; and because every temporal benchmark

is a weighted average of the same three moments, no choice among them resolves this. We therefore report the magnitude as a benchmarked contribution. For the value-weighted WML strategy the abnormal return is 45.3 basis points per month against the mid-month benchmark and 46.6 against all other trading days (Table 6). A separate stock-level design with stock-by-month and calendar-date fixed effects, preserving value weights, gives a larger estimate of 55.4 basis points; its equal-weighted analogue is much smaller, 7.2 basis points, and statistically insignificant (Table IA.11). The contribution is therefore concentrated in the stocks that carry greater weight in the value-weighted strategy, which is the same direction as the liquidity pattern within the loser leg (Table IA.22). The selling, settlement and recovery tests in the main text establish separately that the abnormal return falls where the mechanism predicts and that the stocks bearing the largest relative losses show little subsequent relative recovery.

Why the recovery is not measured against a benchmark for normal loser returns. Measuring how much of the window loss reverses requires a view of what losers would have earned without the selling. A natural candidate is the losers' own return on the mid-month days D, outside PreTOM and the Post window, which captures their usual momentum drift; against it the rebound appears larger than against the market. That benchmark is not clean either. Losers begin to underperform around the eighth trading day of the month, before the window opens (Section IA.20), so the mid-month days already contain part of the month-end selling, and subtracting them removes some of the effect being measured. The market return, which Table 7 uses, ignores the usual drift. Each benchmark is defensible and neither identifies the share of the selling-related loss that reverses. The within-cohort regression below avoids the choice altogether: it compares heavily and lightly sold losers in the same month, so whatever normal loser returns are, they cancel.

Why the recovery is measured within cohorts. Reversal of price pressure is normally identified with an event study. In Coval and Stafford (2007) and Lou (2012) the treated stocks are those held by funds experiencing extreme flows and the event date differs across stocks, so stocks not treated on a given date identify the common component of returns on that date, a date fixed effect removes it, and the abnormal return and its subsequent path are identified without any assumption about the level of the counterfactual rate. Here the event falls on

the same calendar days for every loser, so it is collinear with the calendar: in the notation of equation (IA.1), a date fixed effect would absorb $\mu_W + p_m$ jointly. Other momentum deciles do trade on those dates and do supply same-date comparisons, as in equation (1), but reading such a contrast as the causal effect of month-end selling requires an assumption about how loser and comparison-stock returns would otherwise have varied over the calendar. The same difficulty recurs after the window. A stock can earn a positive return after PreTOM without recovering the loss attributable to the window, because its counterfactual return may also have been positive, and a loser portfolio can rise after month-end without the stocks that fell most during PreTOM catching up with their peers. Section 3.1 of the paper therefore measures recovery within monthly loser cohorts, a comparison that removes the return components common to the cohort and asks whether stocks with larger PreTOM losses subsequently outperform stocks with smaller ones, and so needs no untreated firms.

A common-signal regression and the offset test. The specification the main text uses avoids the ratio entirely. Let $S_{i,m}$ be the stock's average daily seller-initiated imbalance over the window minus its average over the preceding mid-month days. On the same observations and with the same regressors we estimate

$$R^{\text{PreTOM}}_{i,m} = \alpha_m + \beta_{\text{PreTOM}} S_{i,m} + \gamma'_W X_{i,m} + u_{i,m}, \tag{IA.2}$$

$$R^{\text{TOM}}_{i,m} = \delta_m + \beta_{\text{TOM}} S_{i,m} + \gamma'_O X_{i,m} + v_{i,m}, \tag{IA.3}$$

with cohort fixed effects and predetermined characteristics X. A hypothesis that a fraction q of the window loss is offset during TOM is the linear restriction

$$\beta_{\text{TOM}} + q\beta_{\text{PreTOM}} = 0. \tag{IA.4}$$

Because the regressors are identical, regressing $R^{\text{TOM}} + qR^{\text{PreTOM}}$ on them returns exactly $\beta_{\text{TOM}} + q\beta_{\text{PreTOM}}$ with the correct standard error, so the covariance between the initial decline and the subsequent rebound is carried rather than assumed. A significantly negative estimate means less recovery than q requires.

Classical measurement error in S leaves the restriction in equation (IA.4) valid. If S measures true selling pressure with classical error, both β_{PreTOM} and β_{TOM} are attenuated by the same factor, because the same mismeasured regressor appears in both equations on the

same observations, so a restriction that is homogeneous of degree one in the pair continues to hold at the true offset. What attenuation does change is precision: it alters the residual variation and can make rejection harder, so the t-statistic is not invariant and the test may simply lack power. Two assumptions are doing the work and are not testable here: the error in S is classical, uncorrelated with the true signal and with both return errors; and the attenuation factor is common across the two equations, which requires the error structure not to differ between the window and TOM outcomes. Common attenuation cancels in the population coefficient ratio as well, so the advantage of the linear restriction is that it avoids estimating a ratio, not that it is immune to measurement error.

What that comparison identifies. Index losers i within cohort m. Losers in a cohort share the window rates but differ in their exposure to the selling:

$$R^{O}_{i,m} = \mu_{O,m} - \theta_O \, p_{i,m} + \varepsilon^{O}_{i,m}. \tag{IA.5}$$

A cohort fixed effect absorbs $\mu_{O,m}$. If latent pressure $p_{i,m}$ were observed, identifying $-\theta_O$ from its within-cohort variation would additionally require $p_{i,m}$ to be uncorrelated with $\varepsilon^{O}_{i,m}$ after conditioning on the cohort and the included controls. Fixed effects alone do not impose this condition. Latent pressure is not observed here. Section IA.9 uses the observed window return in its place, which measures $p_{i,m}$ with error $\varepsilon^{W}_{i,m}$; the main text instead conditions on incremental selling, as set out below. Under classical measurement error the slope is attenuated toward zero relative to θ_O, so it measures the reversal of the observed relative loss rather than of the pressure inside it. The attenuation factor is the share of the observed window return's variance that is pressure, which is not identified here, and the regression also conditions on the early-month return, so the relevant variances are those of residuals from that control. Section 3.1 of the paper instead conditions directly on incremental selling, which is correlated with $p_{i,m}$, and tests the offset as a linear restriction on the two coefficients rather than as a ratio. Because the selling persists after the window, that sort is not a structural recovery estimator either. A coefficient of -1 corresponds to full recovery of the observed relative loss and zero to none. Over the full Post interval the coefficient on the window return is -0.097, against -0.008 on an equally measured mid-month loss, so the window-specific differential is -0.089.

IA.3 T+1 Settlement Falsification Tests

Two checks confirm that the T+1 settlement result is specific to the boundary days the mechanism predicts and is not the artifact of a slow pre-existing drift. First, *placebo days*: running the identical $\tau-4$ vs. $\tau-3$ DiD but shifted to $\tau-6$ vs. $\tau-7$, two days well inside both the T+2 and T+1 selling windows where the mechanism predicts no relocation of selling pressure, produces a DiD of -1.94 ($t = -0.06$), indistinguishable from zero and in contrast to the $+85.93$ ($t = 2.55$) for the actual May 2024 reform on the same sample (Table IA.1). Second, the *event study* below shows that the within-month $\tau-4$ minus $\tau-3$ differential has no pre-trend in the years before the reform and lifts only after May 2024, supporting the parallel-trends assumption underlying the DiD. The early-month control cancels from the boundary contrast, so adding it changes nothing: the figure here and the one in the main text are the same estimate, and the t-statistics differ only because the main text clusters by month while this table uses robust errors.

Table IA.1: Falsification Tests for T+1 Settlement

Test	DiD	t-stat	p-value	Prediction
Real: Losers, $\tau-4$ vs. $\tau-3$, May 2024	**85.93**	**2.55**	**0.011**	**Nonzero**
Placebo days (losers, May 2024):				
$\tau-6$ vs. $\tau-7$	-1.94	-0.06	0.949	Zero

Notes: Market-adjusted value-weighted loser returns, 1980–2025. For each month we take the difference between the loser returns on the two days compared and regress it on an indicator for months after the T+1 reform; the DiD is that coefficient. Each month contributes one observation, so the two days of a month are not treated as independent, and the real row reproduces the boundary contrast of Table 8. May 2024 is excluded. The placebo comparison uses $\tau-6$ and $\tau-7$ in place of the settlement boundary days. Standard errors are heteroskedasticity-robust.

Event-study pre-trend test. The pooled DiD restricts the data to a single pre/post split at May 28, 2024. A pre-trend in the within-month $\tau-4$ minus $\tau-3$ differential before that date would invalidate the DiD even if the placebo tests above were clean. We test this directly by replacing the T1Reform_t dummy variable with a vector of 12-month event-year bins anchored at May. Bin 0 covers May 2024–April 2025, bin -1 (the omitted reference) covers May 2023–April 2024, bin -2 covers May 2022–April 2023, and so on through bin -8. All months from 1980 through April 2016 are pooled into a single "deep pre" bin. May 2024 is excluded throughout, matching the main-text spec. On the subsample $[\tau-4, \tau-3]$, we estimate

$$r_t = \alpha + \sum_k \alpha_k \, \mathbf{1}[\text{bin}_t = k] + \delta \, \mathbf{1}_{[\tau-4]} + \sum_{k \neq -1} \beta_k \, \mathbf{1}[\text{bin}_t = k] \times \mathbf{1}_{[\tau-4]} + \varepsilon_t,$$

where r_t is the daily market-adjusted value-weighted loser return in basis points. Each β_k is the within-month $\tau-4$ minus $\tau-3$ differential in event-year bin k, relative to the year just before the reform.

Figure IA.1 plots the eleven coefficients with 95% confidence intervals. Standard errors are clustered by calendar month, so that the two boundary days of a month are one monthly observation of the spread rather than two independent draws; this is the same pairing correction the settlement tables apply. The eight pre-event coefficients are scattered around zero, none significant at the 5% level and with no monotone drift toward May 2024; the joint test that all eight equal zero cannot be rejected ($F(8, 550) = 0.76$, $p = 0.64$). Both post-event bins are positive and similar in size: +121.3 basis points for May 2024–April 2025 ($t = 1.97$, $p = 0.05$) and +111.1 for the partial bin May 2025–December 2025 ($t = 1.61$, $p = 0.11$), each on the order of the pooled DiD reported in the main paper. The second bin is not distinguishable from zero under this clustering, so it is the first bin that carries the test; what the second adds is that the point estimate does not revert. The flat pre-period and the discontinuous jump at the reform are what the parallel-trends assumption requires, and neither is contradicted here.

Placebo reform dates. With nineteen post-reform months, we benchmark the boundary statistic against its historical distribution by placing an identically constructed pseudo-reform at each of 305 non-reform months between August 1996 and December 2021, each given the

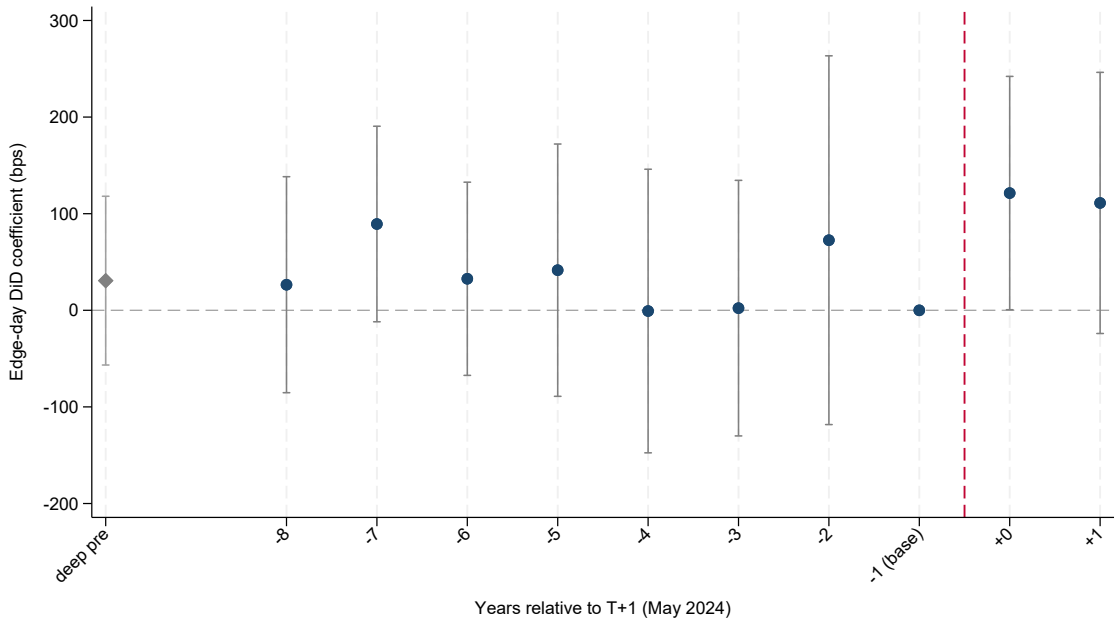


Figure IA.1: Event study of the $\tau-4$ vs. $\tau-3$ DiD. Each point is the within-month $\tau-4$ minus $\tau-3$ differential in market-adjusted value-weighted loser returns (basis points), estimated for the indicated 12-month bin relative to the omitted bin -1 (May 2023–April 2024). Bins are anchored at May, so bin 0 is May 2024–April 2025 and bin -2 is May 2022–April 2023. The deep-pre bin pools 1980 through April 2016. Whiskers are 95% confidence intervals from standard errors clustered by calendar month. May 2024 is excluded. Sample: 1980–2025.

same nineteen-month post period as the actual reform and at least 199 months before it. The pre-period cannot be held equal as well. May 2024, the month in which the change took effect and in which the boundary days straddle it, is excluded throughout, so the post period begins in June 2024 and the actual reform has 532 months before it. A pseudo-reform placed earlier in the sample has fewer, and the realized pre-periods run from 199 to 503 months with a median of 351. Months before August 1996 leave too little pre-period to qualify. The statistic compared against the pseudo-reforms is computed on this construction, by the same estimator applied to each pseudo-reform. The actual shift is more extreme than every one of the 305. We use that tail count as an empirical calibration of how unusual the realized statistic is rather than as a randomization p-value, and the regression estimators of the main text provide the corresponding inference.

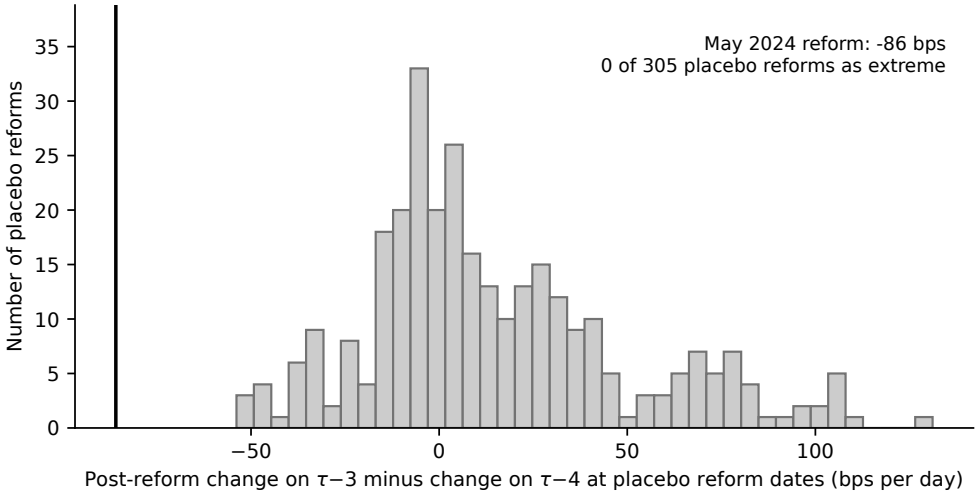


Figure IA.2: Placebo settlement reforms. The distribution of the day-level shift, the post-reform change in the loser market-adjusted return on $\tau-3$ minus the change on $\tau-4$, across 305 placebo reforms placed at each month from August 1996 to December 2021, each with the same nineteen-month post period as the actual reform and at least 199 months before it, from a regression of the daily loser return on day indicators, a post-reform indicator, and their interactions. The vertical line is the May 28, 2024 reform. The actual shift is more negative than every placebo reform.

IA.3.1 What the reform identifies about magnitude

The reform identifies the location of the settlement-sensitive boundary. Its magnitude is less precisely determined, and this subsection records what the design can and cannot say about the size of the migration.

The 7.15-basis-point baseline of Table 2 is not the right comparison for the reform coefficient, because it averages across six window days while the reform moves pressure concentrated at the terminal selling day. Moving x basis points from $\tau-4$ to $\tau-3$ changes their contrast by roughly $2x$, and pre-reform boundary-day effects are on the order of 20 basis points, so the baseline evidence implies a migration contrast of several tens of basis points against a standard error near 34.

The daily pattern looks like a temporal reallocation, with pressure falling at the old boundary and rising on later days. Summing the reform coefficients over $[\tau-6, \tau]$ gives -16.7 basis points of cumulative change and over $[\tau-9, \tau]$ gives -9.8, both with intervals several times their width.

IA.4 Tail Trimming Analysis

Whether the window mean depends on a handful of extreme days can be settled by trimming. We progressively remove the worst days from each window's return distribution and recompute the mean. Figure IA.3 shows the result. PreTOM mean WML is +10.2 bps untrimmed and rises to +17.2 bps after removing the worst 1% of days, so extreme negative days do not generate the positive window mean. The trimming here is one-sided and therefore says nothing about extreme positive days; the two-sided outlier-clean benchmark of Table IA.27 does. In contrast, month-start mean WML flips from −5.5 bps to +4.0 bps after the same 1% trim: about 30 extreme days drive the entire negative month-start average. The two have distinct statistical signatures: the window mean survives the removal of its worst days, while month-start negativity is concentrated in rare tail events.

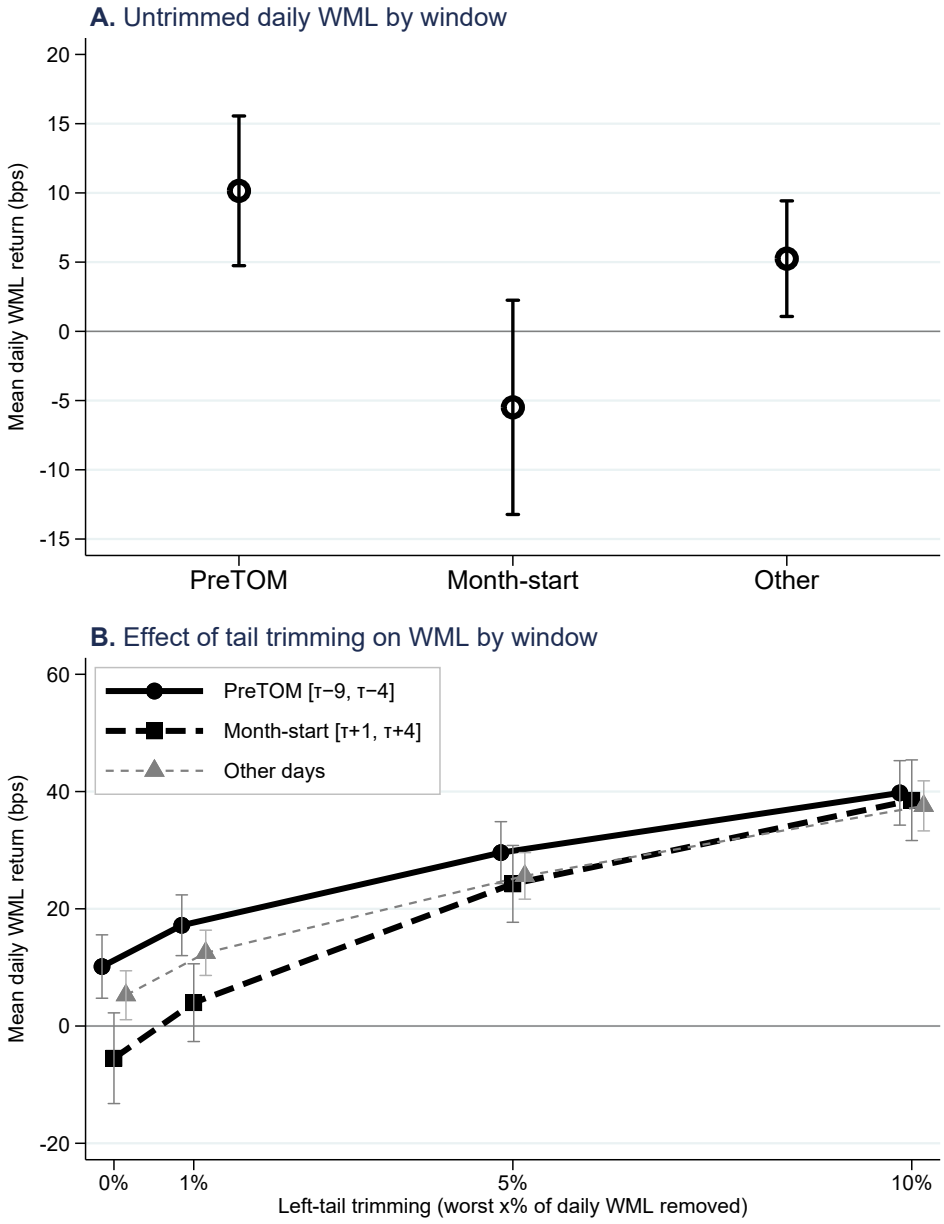


Figure IA.3: Mean daily WML return by calendar window after progressively trimming the worst days from each window's distribution. Panel A: untrimmed mean daily WML by window, with 95% confidence intervals. Panel B: mean daily WML after removing the worst 0, 1, 5 and 10% of days. Windows are PreTOM ($[\tau-9, \tau-4]$), month-start ($[\tau+1, \tau+4]$) and all other days. PreTOM mean WML rises from +10.2 to +17.2 bps after 1% trimming, while month-start flips from −5.5 to +4.0 bps. Sample: 1980–2025, value-weighted.

IA.5 Crash-Day Concentration Methodology

This appendix details the calendar-share construction and proportions z-test summarized in Section 3.3 of the main paper.

The crash-day concentration is displayed against the mean WML profile in Figure 3: mean WML peaks in the window while crash probability peaks at month-start. The month-start concentration holds at every threshold from -100 to -300 basis points (Figure IA.7), and it survives dropping the twenty most extreme WML days ($z = 4.78$), the 1% tails ($z = 4.41$), the 2001–02, 2009 and 2020 episodes ($z = 4.11$), and taking the residual of WML on a trailing 126-day market beta ($z = 4.34$), so the market's own month-start rise interacting with the negative conditional beta of momentum after bear markets does not generate it (Table IA.21). Trimming the tails leaves the window mean intact and flattens month-start (Figure IA.3).

Crash-day classification. We classify a trading day as a crash day if the value-weighted WML return falls below -200 basis points. Robustness at -100 and -300 basis-point thresholds is reported in the figures.

Comparison universes. To prevent the PreTOM and month-start windows from contaminating each other, we compare each window's crash share against its calendar share within a comparison universe that excludes the other window. For PreTOM, the universe is non-month-start trading days (PreTOM days plus other days, excluding $[\tau{+}1, \tau{+}4]$). For month-start, the universe is non-PreTOM trading days (month-start days plus other days, excluding $[\tau{-}9, \tau{-}4]$). The calendar share is the fraction of trading days in the universe that fall in the window of interest. Under the null that crashes are spread proportionally across the calendar, the observed share equals the calendar share.

Test statistic. For each window w and threshold, let $p_{\text{obs}} = n_{\text{crash},w}/n_{\text{crash}}$ be the observed share of crash days in window w and p_{exp} be the calendar share. The standard one-sample proportions z-statistic is

$$z = \frac{p_{\text{obs}} - p_{\text{exp}}}{\sqrt{p_{\text{exp}}(1 - p_{\text{exp}})/n_{\text{crash}}}},$$

two-sided p-values from the standard normal. That variance treats crash days as independent draws, and they are not: crashes arrive in clusters and a handful of months contribute many of them. We therefore also bootstrap over calendar months. The statistic resampled is the difference between the two shares, $p_{\text{obs}} - p_{\text{exp}}$; in each of 5,000 draws we sample months with replacement, let every drawn month contribute all of its days, and recompute both shares. Drawing whole months preserves dependence within a month, which is where the clustering the binomial variance ignores mostly lies. It does not preserve dependence across adjacent months, so an episode spanning several months is broken up by the resampling. The concentration survives: at the -200 basis point threshold the bootstrapped t-statistic is 4.20 on all days, 3.78 excluding the crash episodes, 4.28 dropping the twenty most extreme WML days, 4.01 dropping the most extreme one percent, and 3.97 on the beta residual, against binomial z-statistics of 4.68, 4.11, 4.78, 4.41 and 4.34. Every 95 percent interval excludes zero. Allowing for the clustering costs roughly a tenth of the statistic rather than the result.

Excluded episodes. The robustness reported in the body excludes 21 months, specified as three contiguous periods rather than as individual months: August 2001 through March 2002, January through September 2009, and February through May 2020. The first two span the momentum-crash episodes Daniel and Moskowitz (2016) study and the third the COVID-19 dislocation. Five of the 21 fall in 2001 and sixteen from 2002 on, which is why Table IA.26 reports 531 months against 552 in the full sample, and 259 and 272 against 264 and 288 in the two subperiods. The periods are contiguous windows around known episodes, fixed by calendar dates rather than by realized returns.

IA.6 International Evidence: Methodology

This section details the construction of the international momentum decile portfolios.

Markets and sample period. Daily stock data are from Compustat Global (`comp.g_secd`) for twenty developed markets: Australia, Austria, Belgium, Denmark, Finland, France, Germany, Hong Kong, Ireland, Italy, Japan, the Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland and the United Kingdom. Canada is excluded because Compustat Global covers it thinly; Canadian common stocks sit in the CRSP North America universe. The list is fixed in advance and carries no screen on realized momentum. Japan enters the panel on the same terms as every other market and is reported separately, because prior work finds momentum to be weak there (Fama and French, 2012; Asness et al., 2013); the pooled estimates therefore cover nineteen markets. The panel runs from January 1991 through February 2026, and from 1993 for Hong Kong and Switzerland, where coverage is thin in the early 1990s.

One listing per issuer, and the venue that trades it. Capitalization in Compustat Global (`cshoc`) is reported at the company level, so an issuer with several listings would enter a value-weighted portfolio once per listing at its full capitalization. We keep the single issue Compustat flags as the company's primary row (`g_company.prirow`) and assign each security to the country of the exchange on which that issue trades (`g_security.excntry`), not to the issuer's domicile. The settlement regime a stock trades under is a property of its venue, which is what the European transition in Section 4 exploits.

Currency. Prices are quoted in the currency recorded in `curcdd`, which is not constant within a market. We convert every price to U.S. dollars with Compustat's daily rates (`g_exrt_dly`) before computing any return. Conversion does three things at once. A currency redenomination becomes a pure unit change that cancels against the rate rather than appearing as a return. Capitalizations become comparable across markets, which matters because the breakpoint groups pool small markets. And the portfolio return ends up in the same currency as any dollar benchmark. Because each portfolio and its country market are converted at the same rate, the conversion largely cancels in returns measured relative to the country's own market.

Stock universe and cleaning. We retain common stocks (`tpci='0'`). The total return is built from the dollar price $p^{\$}_{i,t}$, the cumulative split-adjustment factor $a_{i,t}$ (`ajexdi`) and the cumulative dividend reinvestment factor $\tau_{i,t}$ (`trfd`):

$$r_{i,t} = \frac{p^{\$}_{i,t}/a_{i,t} \cdot \tau_{i,t}}{p^{\$}_{i,t-1}/a_{i,t-1} \cdot \tau_{i,t-1}} - 1.$$

We drop observations with a missing return or capitalization, non-positive capitalization, and returns outside $[-0.99, 10]$. Capitalization is the dollar price times `cshoc`. The clean panel holds 93.3 million security-days on 28,965 securities.

Momentum signal. Daily returns are compounded within each security-month, and a security-month is retained only if it contains at least fifteen trading days. The signal is the cumulative log return over eleven months, and it is dated by the month in which the portfolio is *held*: the portfolio held in month h ranks stocks on their returns from $h-12$ through $h-2$, skipping $h-1$. All eleven monthly returns must be observed. This is the 12-2 convention of the U.S. analysis and of Asness et al. (2013).

Decile assignments. Within each country-month, decile cutoffs are computed from stocks at or above the within-country median capitalization at the prior month-end, and all stocks are then assigned to deciles on those cutoffs, so that micro-caps populate the deciles without setting their boundaries. The design is the international analogue of NYSE breakpoints. Five small European markets (Austria, Belgium, Finland, Ireland, Portugal) share one breakpoint group and two small Asia-Pacific markets (New Zealand, Singapore) share another; the remaining thirteen set their own. A group-month is assigned deciles only if it holds at least twenty above-median stocks. Deciles are fixed within the calendar month.

Benchmark. Portfolio returns are value-weighted at the prior month-end. Each leg is benchmarked against its own country's value-weighted market return, built from the same panel and retained on days when at least twenty of that country's stocks trade. A regional benchmark would leave a country-specific component in both legs, which cancels in the long-short but not in either leg on its own, so leg-level estimates for a market that is small within its region would be measured against the wrong thing. The dependent variable in

equation (2) is the day-t value-weighted loser return in country c less that country's market return, in basis points.

Complete months. Relative trading day τ is defined against the last trading day of the calendar month, so a month observed only in part misdates every day in it. We retain a country-month only if the market is observed in some later month, which drops each market's final partial month and nothing else. The rule is mechanical and does not look at returns.

Inference. Standard errors in the pooled regression are clustered by calendar month, which allows loser returns in different countries to be correlated on the same date and is the most conservative of the three schemes we report. Clustering by country-month instead, which treats two countries in the same month as independent, leaves every coefficient unchanged and raises the loser t-statistic from -2.24 to -3.52 and the WML statistic from 2.36 to 4.39. Clustering by date gives -2.36 and 2.59. The leg regressions use 159,159 country-days and the WML regression 162,587, the difference being days on which a country's own market return is unavailable because fewer than twenty of its stocks trade; WML needs no benchmark. Local settlement cycles and institutional practices vary across markets, so a common $[\tau-9, \tau-4]$ window attenuates the pooled estimate wherever the binding boundary falls on a different local trading day.

Full country tables. Tables IA.2 and IA.3 report PreTOM, Rest, and Difference estimates for all 19 countries individually, for both legs (Losers − Market and Winners − Market). Countries are sorted by the loser-leg Difference column.

Table IA.2: International Evidence by Country, Losers minus Own-Country Market (bps/day)

Country	PreTOM	Rest of Month	Difference
Norway	-13.70^{***}	-2.94	-10.76^{***}
	(-4.36)	(-1.37)	(-2.87)
Spain	-9.93^{**}	$+0.44$	-10.37^{**}
	(-2.23)	(0.25)	(-1.97)
Finland	-8.99^{***}	-1.84	-7.14^{*}
	(-2.68)	(-0.89)	(-1.83)
Switzerland	-6.86^{***}	-1.06	-5.80^{**}
	(-2.71)	(-0.67)	(-1.97)
Australia	-9.39^{***}	-3.65^{**}	-5.74^{**}
	(-3.80)	(-2.31)	(-2.15)
UK	-6.83^{***}	-1.48	-5.35^{*}
	(-2.68)	(-0.85)	(-1.75)
Belgium	-6.32^{**}	-1.04	-5.28
	(-2.06)	(-0.44)	(-1.34)
Hong Kong	-7.79^{***}	-2.55	-5.25^{*}
	(-3.27)	(-1.57)	(-1.87)
France	-7.89^{***}	-2.84	-5.05
	(-2.82)	(-1.27)	(-1.34)

Table IA.2: International Evidence by Country, Losers minus Own-Country Market (bps/day) (continued)

Country	PreTOM	Rest of Month	Difference
Denmark	-7.64^{**}	-3.60^{*}	-4.04
	(-2.45)	(-1.69)	(-1.03)
Netherlands	-2.38	$+0.37$	-2.75
	(-0.70)	(0.16)	(-0.67)
Germany	-5.03^{**}	-2.38	-2.64
	(-2.21)	(-1.44)	(-1.01)
Sweden	-3.82	-1.35	-2.46
	(-1.35)	(-0.70)	(-0.77)
Italy	-5.34^{**}	-5.11^{***}	-0.23
	(-2.13)	(-3.47)	(-0.08)
Austria	-3.28	-3.76^{*}	$+0.49$
	(-1.09)	(-1.70)	(0.12)
Singapore	$+0.70$	-1.80	$+2.50$
	(0.35)	(-1.16)	(1.04)
New Zealand	-1.63	-4.92^{**}	$+3.29$
	(-0.39)	(-1.99)	(0.72)
Ireland	$+2.57$	-1.14	$+3.71$
	(0.35)	(-0.20)	(0.40)
Portugal	$+2.50$	-4.12^{*}	$+6.62$
	(0.47)	(-1.76)	(1.14)
Pooled	-5.52^{***}	-2.34^{**}	-3.18^{**}
	(-4.24)	(-2.46)	(-2.24)

Notes: Daily value-weighted loser returns in U.S. dollars above each country's own value-weighted market, in basis points, 1991–February 2026; Hong Kong and Switzerland begin in 1993. PreTOM is local trading days $[\tau-9, \tau-4]$; Rest comprises other days. Countries are ordered by PreTOM−Rest. Parentheses report t-statistics with standard errors clustered by calendar month. The pooled row includes country fixed effects and excludes Japan. Portfolio construction is described in Section IA.6. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.3: International Evidence by Country, Winners minus Own-Country Market (bps/day)

Country	PreTOM	Rest of Month	Difference
Belgium	+0.49	+3.17*	−2.67
	(0.19)	(1.95)	(−0.84)
Austria	+2.31	+4.22*	−1.91
	(0.70)	(1.90)	(−0.44)
Hong Kong	−0.01	+1.67	−1.68
	(−0.00)	(0.91)	(−0.45)
Australia	+6.70***	+7.10***	−0.40
	(2.64)	(3.76)	(−0.13)
Italy	+2.26	+2.33**	−0.07
	(1.17)	(2.15)	(−0.03)
New Zealand	+2.45	+2.05	+0.40
	(0.55)	(1.12)	(0.09)
Norway	+5.41*	+4.86***	+0.55
	(1.75)	(2.58)	(0.15)
UK	+3.75**	+2.68**	+1.08
	(2.19)	(2.19)	(0.49)
Denmark	+5.27**	+3.41**	+1.86
	(2.05)	(2.28)	(0.59)

Table IA.3: International Evidence by Country, Winners minus Own-Country Market (bps/day) (continued)

Country	PreTOM	Rest of Month	Difference
Netherlands	+2.98	+1.09	+1.90
	(1.35)	(0.76)	(0.71)
Germany	+5.83***	+3.04**	+2.78
	(2.69)	(2.32)	(1.10)
Portugal	+6.69*	+3.87	+2.82
	(1.81)	(1.50)	(0.62)
Finland	+5.70**	+2.73	+2.97
	(2.02)	(1.62)	(0.93)
France	+5.41***	+2.21*	+3.20
	(2.82)	(1.83)	(1.32)
Spain	+4.54**	+0.20	+4.34*
	(2.46)	(0.16)	(1.85)
Singapore	+5.55**	+0.75	+4.80
	(2.32)	(0.41)	(1.47)
Switzerland	+6.10***	+1.18	+4.92**
	(3.04)	(0.94)	(1.96)
Sweden	+5.88**	+0.31	+5.57*
	(2.31)	(0.20)	(1.81)
Ireland	+7.63	−5.25	+12.88
	(1.10)	(−1.27)	(1.49)
Pooled	+4.40***	+2.31***	+2.09
	(3.84)	(3.26)	(1.56)

Notes: Daily value-weighted winner returns in U.S. dollars above each country's own value-weighted market, in basis points, 1991–February 2026; Hong Kong and Switzerland begin in 1993. PreTOM is local trading days $[\tau-9, \tau-4]$; Rest comprises other days. Countries are ordered by PreTOM−Rest. Parentheses report t-statistics with standard errors clustered by calendar month. The pooled row includes country fixed effects and excludes Japan. Portfolio construction is described in Section IA.6. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.7 Historical PreTOM Concentration

Table IA.4 reports daily PreTOM and Rest mean returns by historical subperiod for D1 (Panel A) and WML (Panel B). Each panel reports both value-weighted and equal-weighted returns, using the period boundaries shown in the table. PreTOM days are 29% of the trading month. If daily returns were generated uniformly across the month, PreTOM and Rest mean returns would coincide up to sampling error; any systematic gap between them reflects a calendar-time concentration in the underlying return process.

The PreTOM and Rest-of-month rates are similar in the first half of the century, and the differential builds from mid-century onward, coinciding with the scaling of institutional ownership in U.S. equities. Holding the universe fixed at NYSE, so that the comparison does not also move the set of listed stocks, the loser window differential is +0.54 basis points per day in 1927–1951, −1.69 in 1952–1961, −4.73 in 1962–1979, −5.10 in 1980–2001 and −7.06 after 2002. The NYSE-only comparison uses the subperiods listed here. The emergence is gradual rather than dated to a single year.

Table IA.4: Subperiod Analysis of PreTOM versus Rest, D1 (Loser − Mkt) and WML

Panel A: D1 (Loser − Mkt) returns (bps/day)

	Value-weighted			Equal-weighted			
Period	PreTOM	Rest	PreTOM − Rest	PreTOM	Rest	PreTOM − Rest	Inst. Own.
1927–1940	−5.06	−4.72	−0.35	19.69***	23.33***	−3.64	< 10%
	(−0.83)	(−1.49)	(−0.05)	(2.98)	(5.24)	(−0.52)	
1940–1962	−2.35*	−2.19**	−0.16	2.22	3.58**	−1.36	< 10%
	(−1.71)	(−2.36)	(−0.10)	(1.31)	(2.43)	(−0.71)	
1962–1980	−8.01***	−2.56**	−5.45***	−1.66	4.56**	−6.21***	15–30%
	(−4.09)	(−2.02)	(−2.78)	(−0.76)	(2.42)	(−2.99)	
1980–2001	−9.65***	−4.14**	−5.51*	8.46***	11.46***	−3.00	30–55%
	(−4.14)	(−2.32)	(−1.92)	(3.34)	(4.93)	(−1.08)	
2002–2025	−6.46*	2.28	−8.73**	3.08	3.89	−0.80	55–72%
	(−1.83)	(0.97)	(−2.02)	(0.91)	(1.62)	(−0.21)	

Panel B: WML (Winner − Loser) returns (bps/day)

	Value-weighted			Equal-weighted			
Period	PreTOM	Rest	PreTOM − Rest	PreTOM	Rest	PreTOM − Rest	Inst. Own.
1927–1940	2.25	9.17**	−6.92	−13.59**	−11.73***	−1.87	< 10%
	(0.28)	(2.16)	(−0.84)	(−1.96)	(−2.76)	(−0.27)	
1940–1962	5.75***	4.85***	0.90	3.58*	1.50	2.08	< 10%
	(2.76)	(3.47)	(0.36)	(1.82)	(0.99)	(0.90)	
1962–1980	10.74***	8.10***	2.64	6.80***	2.61	4.19*	15–30%
	(3.73)	(4.31)	(0.87)	(3.19)	(1.45)	(1.74)	
1980–2001	10.24***	7.01***	3.23	−6.01***	−4.65**	−1.36	30–55%
	(3.29)	(2.93)	(0.84)	(−2.76)	(−2.24)	(−0.54)	
2002–2025	10.06**	−1.95	12.02**	1.08	−3.16	4.23	55–72%
	(2.19)	(−0.67)	(2.13)	(0.34)	(−1.37)	(1.15)	

Notes: Daily mean returns in basis points per day, by historical subperiod and weighting scheme. Panel A reports the D1 (loser − Mkt) portfolio return and Panel B the WML (winner − loser) excess return. PreTOM is the trading-day window $[\tau-9, \tau-4]$ before month-end (τ is the last trading day); Rest is all other trading days. As in Table 5, daily returns are first averaged separately within the PreTOM and Rest windows of each month, and every mean and t-statistic is computed across these monthly averages. The PreTOM − Rest column reports the mean of the within-month difference between the two averages. Every column carries its own asterisks, taken from the t-statistic printed directly beneath that entry, so a level star refers to the level and a difference star to the difference. Aggregate institutional ownership figures are approximate ranges from Blume and Keim (2012) and the NYSE Fact Book. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.8 Mutual Fund Cash Buffers

Figure IA.4 plots the cash share of active U.S. equity mutual funds from the CRSP Survivor-Bias-Free Mutual Fund Database, excluding index funds and exchange-traded funds. The median fund held 4.5 percent of its assets in cash in 1992–1997, 2.1 percent in 2003–2015 and 1.1 percent in 2016–2024. The asset-weighted mean fell from 7.0 to 1.9 percent over the same years. The fall in 2016 is a cut by continuing funds over three quarters, not a change in reporting, and it coincides with the adoption of the SEC's fund liquidity rule (Rule 22e-4). Funds use cash to meet redemptions rather than trade their holdings, and these buffers are too small to prevent fund flows from moving prices (Chernenko and Sunderam, 2016, 2020). A thinner buffer leaves more of each month's redemptions to be met by selling stocks. Credit lines, interfund lending and index futures also supply liquidity, so the decline in cash overstates the decline in funds' total liquidity.

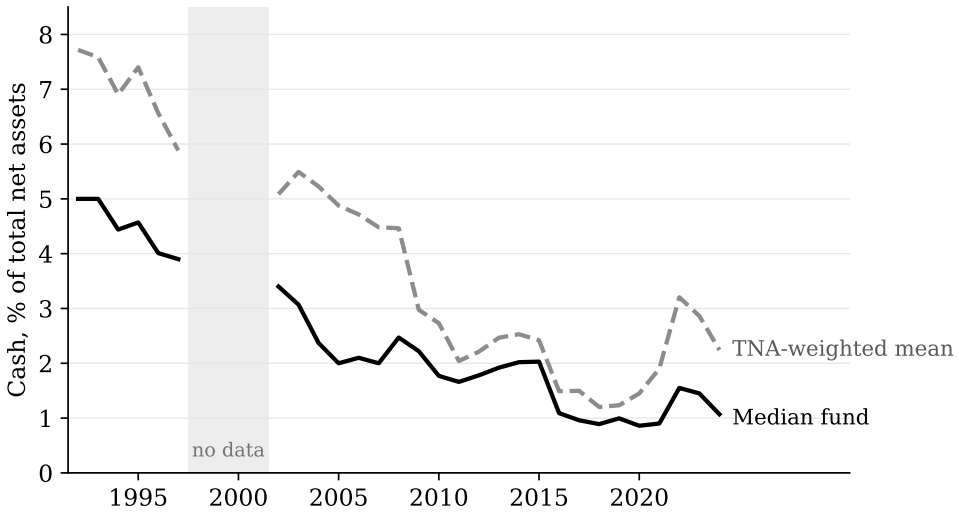


Figure IA.4: Cash holdings of active U.S. equity mutual funds, percent of total net assets. The solid line is the median fund and the dashed line the asset-weighted mean, by year. CRSP reports almost no fund cash data for 1998–2001, which are omitted.

IA.9 Recovery of Selling-Associated Losses: Specifications

The main text tests whether the turn-of-month rebound offsets the losses associated with heavier PreTOM selling, using the window and post-window returns of the same stocks related to the same selling signal. Table IA.5 reports that regression and the offset tests across the three samples. This section collects two complementary pieces of evidence: a return-on-return regression within the same cohorts, which asks whether the losers that fell furthest catch up with the rest of their cohort, and portfolio sorts that compare the most and least heavily sold losers directly.

Table IA.5: Recovery of Selling-Associated PreTOM Losses

	Cap-weighted	Unweighted	Restricted
PreTOM window return	-131.6^{***}	-305.1^{***}	-155.1^{***}
	(-13.25)	(-34.90)	(-18.68)
TOM return	$+8.7$	$+23.0^{***}$	$+5.0$
	(0.75)	(3.93)	(0.67)
Full recovery test: $\beta_{\text{TOM}} + \beta_{\text{PreTOM}} = 0$			
Full recovery ($q = 1$)	-122.9^{***}	-282.2^{***}	-150.1^{***}
	(-8.41)	(-31.46)	(-12.77)
Share recovered, 95% interval	$[-11, 24]\%$	$[4, 11]\%$	$[-6, 13]\%$
Loser stock-months	132,637	132,637	24,916

Notes: Coefficients on S, the stock's average daily seller-initiated imbalance during PreTOM minus its average over the preceding mid-month window, scaled to one within-cohort standard deviation of S and reported in basis points. The first row is the cumulative market-adjusted return over the window, the second the cumulative return over the seven trading days of TOM. The recovery test regresses $R^{\text{TOM}} + R^{\text{PreTOM}}$ on the same regressors; because the regressors are identical the coefficient is exactly $\beta_{\text{TOM}} + \beta_{\text{PreTOM}}$ with the correct standard error, so the covariance between the initial decline and the subsequent rebound is carried rather than assumed. A significantly negative value means the rebound falls short of the window loss. The share recovered is the fraction q of the window loss offset during TOM; its 95% interval collects every q for which the offset test $\beta_{\text{TOM}} + q\beta_{\text{PreTOM}} = 0$ is not rejected at the 5% level. All specifications include cohort fixed effects and predetermined controls for momentum, recency of loss, size, volatility, market beta, bid-ask spread, dollar volume and the mid-month selling rate. Cap-weighted weights observations by market capitalization; Restricted keeps losers in the largest third of the decile that a mutual fund held at the preceding Thomson S12 report date. The standard deviation of S differs across columns, so levels are comparable within a column and not across. Sample 2003–2022. t-statistics in parentheses, two-way clustered by stock and by the calendar month in which the outcome window ends. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A return-on-return regression. Rather than relating both returns to selling, one can relate the post-window return directly to the window return. Within monthly loser cohorts we estimate

$$R_{i,m}^{\text{Post}} = \beta_1 R_{i,m}^{\text{PreTOM}} + \beta_2 R_{i,m}^{\text{Early}} + \mu_m + \varepsilon_{i,m}, \tag{IA.6}$$

where μ_m is a cohort fixed effect and R^{Early} is the cumulative market-adjusted return over Early, the first six trading days of Mid in the cohort month.[4] A coefficient of -1 on the window return denotes full recovery of the observed relative loss, zero denotes none. Over the seven trading days immediately following PreTOM it is -0.097, with a 95 percent interval of $[-0.109, -0.085]$, against -0.008 on the Early return, a difference of -0.089 (Table IA.6, Panel A). A stock with a 100-basis-point larger relative window loss therefore recovers about 10 basis points relative to its cohort, on 476,736 loser stock-months in 550 cohorts.

This estimator answers a different question, whether the losers that fell furthest catch up with the rest of their cohort, and it carries a defect the selling-based specification does not: its regressor and its outcome share the closing price of the window's last day, so a one-day microstructure reversal loads on the measured give-back. Opening a single-day gap between the two roughly halves it, from -0.089 to -0.039, and the interaction with the log bid-ask spread that flags bid-ask bounce falls from -0.026 ($t = -3.6$) to -0.005 ($t = -0.8$). The selling-based specification shares no price between regressor and outcome and is not exposed to this channel.

[4]Early is the first six trading days at position $\tau+4$ or later, so in every month of the sample its days fall strictly before PreTOM begins. A cohort month is dropped rather than allowed to overlap when Mid supplies fewer than six such days, which excludes November 1980 and September 2001.

Table IA.6: The Window Return and Subsequent Returns Within Monthly Loser Cohorts

Panel A: Recovery of PreTOM losses within loser cohorts, U.S. losers, 1980–2025				
Outcome horizon	PreTOM (β_1)	Placebo (β_2)	$\beta_1 - \beta_2$	N
Full Post $[\tau{-}3, \tau{+}3]$	-0.097^{***}	-0.008^{*}	-0.089^{***}	476,736
	(-15.9)	(-1.9)	(-14.8)	
Days $[\tau{+}1, \tau{+}3]$	-0.012^{***}	$+0.003$	-0.015^{***}	476,736
	(-3.6)	(1.1)	(-4.2)	
First month after	-0.010	$+0.018^{**}$	-0.028^{***}	476,736
month-end $[\tau{+}1, \tau{+}21]$	(-1.3)	(2.5)	(-3.6)	
Months 2 to 3	$+0.059^{***}$	$+0.055^{***}$	$+0.004$	469,800
	(6.2)	(5.8)	(0.4)	
Months 7 to 12	$+0.145^{***}$	$+0.133^{***}$	$+0.012$	433,433
	(9.7)	(8.6)	(0.8)	

Panel B: Most-sold minus least-sold losers, 2003–2022				
	Return (bps)	t	Net selling pressure	t
PreTOM $[\tau{-}9, \tau{-}4]$	-640	(sort)	$+0.330$	(sort)
Days $[\tau{+}1, \tau{+}3]$	$+17$	1.7	$+0.076$	28
Days $[\tau{+}4, \tau{+}21]$	-44	-2.0	$+0.070$	28
Months 2 to 3	-154	-4.5		
Months 4 to 6	-251	-6.2		

Notes: Panel A estimates equation (IA.6) within monthly loser cohorts, with cohort fixed effects. Outcomes and regressors are cumulative market-adjusted returns. The placebo (β_2) uses Early, the first six Mid trading days, in place of the PreTOM window; November 1980 and September 2001 are omitted because fewer than six are available. Returns include delisting adjustments and end at delisting. The three estimate columns of Panel A come from the same fit and the same covariance matrix. Parentheses report t-statistics with standard errors two-way clustered by stock and by the calendar month in which the outcome window ends. For the two shortest horizons, consecutive cohorts' outcome windows do not overlap and clustering on the cohort gives the same statistics to within 0.3; the longer horizons overlap across cohorts, where the two-way estimator is the relevant one. Panel B reports equal-weighted most-sold minus least-sold tercile spreads, sorting on PreTOM net selling pressure, (sell volume − buy volume)/total volume. Its t-statistics use 230 monthly cohort spreads. The PreTOM row is the sorting window itself, so no t-statistic is reported for it: the spread is large there by construction, and the cells are marked (sort). Construction and recovery interpretation are in Section IA.2; additional specifications are in Tables IA.24–IA.25. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. $\beta_1 = -1$ is full recovery of the window move, $\beta_1 = 0$ none; -0.097 is about 10% recovered.

IA.10 Recovery Response under Equal Weighting

The main text reports the cumulative return response to incremental PreTOM selling under value weighting. Figure IA.5 repeats those regressions equal weighted. The short-horizon result is unchanged. The window response is −305.1 basis points per standard deviation of incremental selling, and only 23.0 of it has returned by the end of TOM. Equal weighted, the losses also last longer. The path recovers gradually and still leaves 211.6 basis points outstanding at twelve months, with every interval excluding zero.

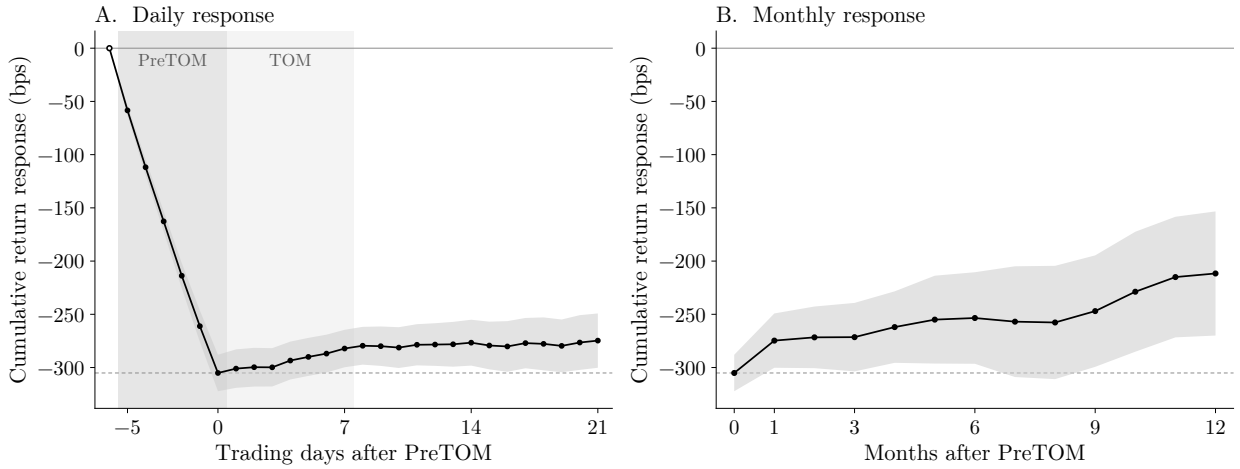


Figure IA.5: Cumulative return response to incremental PreTOM selling, equal weighted. The equal-weighted companion to Figure 5, which is value weighted; construction, clock, controls, sample and clustering are otherwise identical. Each point is the coefficient on S, the average daily net selling pressure over the PreTOM window minus the average over the preceding Mid days, in a regression of the cumulative market-adjusted return accumulated from immediately before the window through that horizon, scaled to one within-cohort standard deviation of S. Day 0 is the final PreTOM trading day and a month is 21 trading days. The solid horizontal line marks full recovery of the window move and the dashed line the cumulative response on the last window day, −305.1 basis points. The dashed line is itself an estimated quantity rather than a fixed benchmark, so whether a plotted band crosses it is not a test; the formal comparisons use the joint covariance of the two coefficients and are reported separately. Shaded bands are 95 percent intervals, two-way clustered by stock and by the calendar month in which the outcome window ends.

IA.11 Long-Horizon Path of New Losers

Figure IA.6 follows stocks that newly enter the loser decile, those in the bottom decile in month m but not in month $m-1$, for twelve months from the first PreTOM day of the entry month, whatever their later decile. Each of the 552 monthly cohorts from 1980 to 2025 is value weighted by market capitalization at the start of the window, and returns are measured relative to the market. CRSP delisting returns are included, with -30 percent for performance-related delistings whose return is missing (Shumway, 1997).

New entrants lose -35.6 basis points during the window ($t = -2.49$). They keep underperforming for about 4 months, to a cumulative -106 basis points, then rise, returning to their pre-window level after about 9 months. The path combines the window loss with ordinary loser dynamics. The effect of the selling itself comes from the within-cohort comparison in Section 3.1 of the paper, and there the losers sold hardest keep falling relative to those sold least for six months after the window (Table IA.6, Panel B). Figure 4 of the paper shows the first four months of the same cohorts day by day, against new entrants to the winner decile.

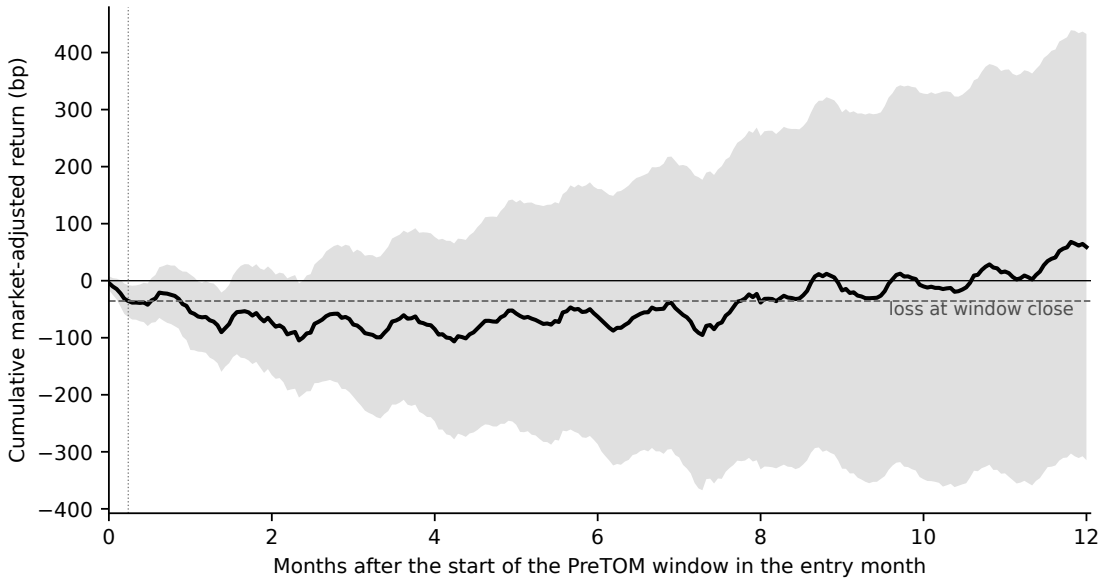


Figure IA.6: Cumulative market-adjusted return of new entrants to the loser decile, from the first PreTOM day of the entry month, in basis points. Value-weighted frozen cohorts, 1980–2025, with CRSP delisting returns. The dashed line marks the loss at the close of the window. The shaded band is a 95 percent confidence interval from Newey–West standard errors across overlapping monthly cohorts, with lag equal to the horizon in months.

IA.12 Same-Weekday Momentum and the Window

This section reports three tests separating the window from the same-weekday continuation of Da and Zhang (2025). Their decomposition replicates on our sample before anything is tested against it. Same-weekday cells account for 69.3 percent of the Fama–MacBeth momentum coefficient equal weighted, against the 70 percent they report, on a universe matching theirs, namely price at least five dollars at the prior month end and capitalization above the NYSE tenth percentile.

Table IA.7 reports the first test. The second splits their daily-rebalanced same-weekday strategy by calendar position. It earns 6.79 basis points a day equal weighted ($t = 7.06$) and takes 33.0 percent of its market-adjusted return on the window's 28.6 percent of days. The third sorts losers within each cohort on their own average formation-period return on the weekday in question and finds no difference across that sort in the window loss, with a top-minus-bottom difference of -1.02 ($t = -0.38$).

Table IA.7: The Window Differential by Weekday

Leg and universe	Mon	Tue	Wed	Thu	Fri	Mean	Wald p
D1, all exchanges	-12.89^{***}	-8.80	-9.07^{**}	0.56	-9.39^{**}	-8.46	0.4471
	(4.16)	(6.01)	(3.97)	(5.87)	(3.90)		
D1, NYSE	-14.99^{***}	-5.75	-5.16	-1.05	-8.80	-7.61	0.2347
	(3.99)	(6.59)	(4.72)	(5.98)	(5.49)		
D10, all exchanges	7.19^{*}	9.93^{***}	-1.27	-3.66	-6.52^{*}	1.17	0.0442
	(3.80)	(3.39)	(4.16)	(3.61)	(3.48)		
D10, NYSE	7.26^{*}	11.06^{***}	0.02	-6.64^{*}	-3.93	2.02	0.0242
	(3.80)	(3.93)	(3.39)	(3.57)	(3.33)		
WML, all exchanges	20.07^{***}	18.73^{**}	7.80	-4.22	2.87	9.63	0.1452
	(6.40)	(7.35)	(6.67)	(7.45)	(5.87)		
WML, NYSE	22.24^{***}	16.81^{*}	5.17	-5.58	4.88	9.64	0.0458
	(6.11)	(8.79)	(6.99)	(7.52)	(6.68)		

Notes: PreTOM−Rest differentials in daily value-weighted market-adjusted returns, estimated separately by weekday, 1980–2025, in basis points. Inference uses Newey–West standard errors with twelve lags on the event-month series. Mean is the five-weekday average for months with all five weekdays observed. The Wald statistic tests equality of the five weekday differentials on this balanced sample. Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.13 Is the Window Selling Informed?

Table IA.8 asks whether the selling that is incremental to the window forecasts subsequent news. Ordinary institutional selling need not be uninformed, and the table does not establish that it is; what it bounds is the information content of the window increment, which is the quantity the mechanism turns on. The increment is estimated close to zero on every outcome with power, and its point estimates are positive rather than negative. A six-day block in mid-month produces nothing, so nothing about carving six days out of a month mechanically generates a nonzero increment.

For each bottom-decile stock i and month m in 2003–2022, let $S^{\text{in}}_{i,m}$ be the average daily net selling pressure, $\text{NetSell}_{i,t}$ of equation (7) in the paper, over the PreTOM window $[\tau_m{-}9, \tau_m{-}4]$, and let $S^{\text{out}}_{i,m}$ be its average over the other trading days of the same calendar month. A stock-month enters if at least three window days and five other days are observed. We estimate

$$\text{News}_{i,m+1} = a_m + b_1\, S^{\text{out}}_{i,m} + b_2 \left(S^{\text{in}}_{i,m} - S^{\text{out}}_{i,m}\right) + \gamma' X_{i,m} + \varepsilon_{i,m}, \tag{IA.7}$$

where a_m is a calendar-month fixed effect, so each loser is compared only with losers of the same month. b_1 is the news content of selling outside the PreTOM window and b_2 that of the window increment. $X_{i,m}$ contains the log of market capitalization, the log of turnover, the stock's market-adjusted return over the window, and the log of the book-to-market ratio. $\text{News}_{i,m+1}$ is measured at the first earnings announcement after the end of month m and is winsorized at the 1st and 99th percentiles. The standardized earnings surprise is the seasonal change in split-adjusted quarterly earnings per share, scaled by its standard deviation over the previous eight quarters. The announcement return is the stock's return minus the CRSP value-weighted market return over trading days -1 to $+1$ around the announcement, in basis points. Standard errors are clustered by stock and by month. Because $b_1 S^{\text{out}} + b_2(S^{\text{in}} - S^{\text{out}}) = b_2 S^{\text{in}} + (b_1 - b_2) S^{\text{out}}$, the test of $b_1 = b_2$ asks whether a unit of window selling carries the same news as a unit of selling outside the PreTOM window. The placebo replaces the window with the six trading days $[\tau_m{-}15, \tau_m{-}10]$ in mid-month and is otherwise identical.

Table IA.8: Does the Incremental Window Selling Forecast Subsequent News?

Outcome	N	Ordinary	Increment	$p(b_1{=}b_2)$	Per s.d.
Panel A: the window, $[\tau{-}9, \tau{-}4]$					
Next earnings surprise	124,562	-0.065^{*}	0.026	0.004	0.005
		(0.038)	(0.020)		$[-0.002, 0.012]$
Next announcement return	127,255	-64.190^{*}	2.047	0.042	0.372
		(36.934)	(20.105)		$[-6.783, 7.526]$
Analyst revision, three months	50,933	-36.254	22.495	0.233	2.942
		(54.390)	(27.929)		$[-4.218, 10.102]$
Panel B: placebo, a six-day block in mid-month					
Next earnings surprise	124,360	-0.071^{*}	-0.024	0.159	
		(0.038)	(0.022)		
Next announcement return	127,053	-66.697^{*}	-27.910	0.260	
		(36.736)	(19.775)		
Analyst revision, three months	50,922	-32.196	-25.391	0.876	
		(54.366)	(32.528)		

Notes: Bottom-decile stock-months in the 2003–2022 WRDS Intraday Indicators sample. Ordinary is the coefficient on non-PreTOM net selling pressure; Increment is the coefficient on PreTOM pressure minus non-PreTOM pressure. Both enter jointly with cohort fixed effects. Standard errors are two-way clustered by stock and month. Per s.d. scales the Increment coefficient by its regressor's standard deviation, in the outcome's units; the interval beneath is 95%. Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.14 Window Dressing Falsification

Brown (2017) documents that momentum returns concentrate in the third month of calendar quarters, attributing the pattern to tax-loss selling and window dressing by delegated managers.

If window dressing is the primary driver of the pre-month-end loser effect, the pattern should amplify in December and at quarter-ends. Our evidence points in the opposite direction. The loser window effect remains present outside quarter-end months and outside December, and the quarter-end interaction is estimated too imprecisely to speak to amplification either way.

Three tests address the window dressing and tax-loss alternatives directly, all reported in Table IA.9. First, restricting the sample to non-quarter-end months, where window dressing is least relevant (column 1), the Loser $\times$ PreTOM coefficient is -7.5 bps ($t = -2.66$), virtually identical to the full-sample estimate. Second, restricting the sample to non-December months, where tax-loss harvesting is least relevant (column 2), the Loser $\times$ PreTOM coefficient is -7.03 bps ($t = -2.86$), again virtually identical to the full-sample estimate of -7.15 bps. The loser window effect therefore remains present outside quarter-end months and outside December. Third, interacting Loser $\times$ PreTOM with a quarter-end indicator (column 3) produces a triple interaction that is estimated imprecisely, consistent with a monthly liquidity cycle rather than quarterly reporting incentives.

Table IA.9: Window Dressing and Tax-Loss Falsification

	(1) Non-QE months	(2) Non-December months	(3) Full sample, QE interaction	(4) Full sample baseline
Loser	4.132***	2.756**	4.039***	2.609**
	(2.68)	(2.05)	(2.62)	(2.04)
Loser × PreTOM	−7.523***	−7.027***	−7.496***	−7.151***
	(−2.66)	(−2.86)	(−2.65)	(−3.08)
Loser × QtrEnd			−4.249	
			(−1.48)	
Loser × PreTOM × QtrEnd			0.990	
			(0.18)	
Constant	−0.017	−0.078	−0.162	−0.162
	(−0.12)	(−0.63)	(−1.41)	(−1.39)
Fixed effects	Stock, Date	Stock, Date	Stock, Date	Stock, Date
Observations	35,417,562	48,853,202	53,324,145	53,324,145
Within R^2	< 0.0001	< 0.0001	< 0.0001	< 0.0001

Notes: Dependent variable is the daily market-adjusted return ($r_{i,t} - r_t^m$, in excess of the value-weighted CRSP market return), in basis points. Returns are value-weighted using lagged market-capitalization weights. Column (1) restricts the sample to non-quarter-end months (Jan, Feb, Apr, May, Jul, Aug, Oct, Nov); window dressing would be least relevant in these months, yet the Loser × PreTOM coefficient is essentially identical to the full-sample estimate. Column (2) restricts the sample to non-December months; tax-loss harvesting is most concentrated in December, and excluding it leaves the Loser × PreTOM coefficient essentially unchanged. Column (3) estimates the full sample with quarter-end interactions; QtrEnd equals one in March, June, September, December. The triple interaction Loser × PreTOM × QtrEnd is small and statistically indistinguishable from zero, indicating no quarter-end amplification. Column (4) reproduces the full-sample baseline from Table 2 of the main paper for reference. All specifications include stock and date fixed effects. t-statistics (in parentheses) are computed from standard errors two-way clustered by stock and date. Momentum deciles use NYSE breakpoints applied to all NYSE, AMEX, and NASDAQ stocks in CRSP; assignments are held constant within each calendar month. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sample: 1980–2025.

IA.15 52-Week-High Sort: Additional Evidence

This section reports additional evidence on the 52-week-high sort discussed in Section 6 of the main paper. Figure 8 compares the share of non-paying firms across deciles under the two sorts.

The characteristic. The sort uses `prc_highprc_252d` from Jensen et al. (2023), the split-adjusted price divided by its maximum over the preceding 252 trading days, dated at the end of the month before the returns it sorts. The primary measure uses split-adjusted prices; the alternative cumulates total returns, so its running maximum is a total-return high. The two rank stocks almost the same way, with a rank correlation of 0.97 and the same quintile assigned to 88 percent of loser stock-months, but reinvested dividends raise a payer's index about one percent a year above its price, which moves payers toward the near-the-high end of a sort whose far end is 89 percent non-payers.

Measure and coverage. The characteristic is merged onto our panel, so the larger universe it comes from cannot add stocks to ours; 86.7 percent of our loser stock-days find a value. The ones that do not are small, with a median capitalization of $3.6 million against $45.9 million, and early: 39.5 percent of the 1980s are unmatched against 0.1 percent from 2000 onward. Since coverage moves with the measure, we compare the two constructions on the rows where both characteristics exist. There the window spread is 7.66 basis points per day on our own construction and 8.76 on the JKP measure, with window-minus-rest differentials of 5.03 and 6.23. Neither the choice of measure nor the coverage difference changes the conclusion.

Table IA.10: Intraday and Overnight Decomposition of the Window Loss

Leg	Session	PreTOM	Rest	PreTOM − Rest
Losers − Mkt	Intraday (open to close)	-10.49^{***}	-3.46^{**}	-7.03^{***}
		(2.39)	(1.54)	(2.70)
	Overnight (close to open)	$+2.67$	$+4.19^{***}$	-1.52
		(1.84)	(1.02)	(1.88)
	Close to close	-8.61^{***}	$+0.16$	-8.76^{**}
		(2.90)	(1.98)	(3.54)
Winners − Mkt	Intraday (open to close)	-3.06^{**}	-4.56^{***}	$+1.50$
		(1.45)	(0.99)	(1.80)
	Overnight (close to open)	$+6.14^{***}$	$+6.26^{***}$	-0.12
		(0.76)	(0.49)	(0.87)
	Close to close	$+3.02^{*}$	$+1.62$	$+1.40$
		(1.67)	(1.09)	(2.07)

Notes: Daily value-weighted returns of the bottom and top momentum deciles, 1992–2024, in basis points per day. Returns are measured relative to the value-weighted return of the eligible stock universe in the same session. Intraday returns run from opening to closing prices; overnight returns are recovered from total daily returns by compounding (Lou et al., 2019). Because the two sessions compound rather than add, they need not sum to the close-to-close row. PreTOM is $[\tau-9, \tau-4]$ and Rest all other days. Parentheses report standard errors computed across months. Eligibility screens and price treatment are in Section IA.1. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.11: The Window Excess Within Stock-Month and Within Date

Weights	Comparison days	Loser × PreTOM	Winner × PreTOM	Loser contribution	Winner contribution	WML excess
Value-weighted	all other days	-8.87^{***}	$+0.36$	$+53.2^{***}$	$+2.2$	$+55.4^{***}$
		(2.27)	(1.93)	(13.6)	(11.6)	(19.4)
Equal-weighted	all other days	-3.08^{**}	-1.87	$+18.5^{**}$	-11.2	$+7.2$
		(1.30)	(1.51)	(7.8)	(9.0)	(11.2)

Notes: Daily stock returns, 1980–2025, regressed on Loser × PreTOM and Winner × PreTOM with stock-month and date fixed effects. The sample contains 53,324,149 stock-days and 2,547,524 stock-months. Comparison days are all non-PreTOM days. Value weights use lagged capitalization within decile-date. Coefficients are in basis points per day; monthly contributions multiply by six and reverse the loser coefficient's sign. Abnormal WML is six times the winner-minus-loser coefficient difference. Parentheses report standard errors clustered by date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.12: The PreTOM Window's Abnormal Contribution to Momentum: CAPM-Adjusted Loser Contribution

Benchmark	Abnormal WML		Loser abnormal contribution		Loser abnormal contribution CAPM-adjusted	
	bps/ month (1)	Share of WML (2)	bps/ month (3)	Share of abnormal WML (4)	bps/ month (5)	Share of abnormal WML (6)
Panel A: Portfolio returns, temporal benchmark						
All other days (Rest)	+46.6**	48%	+42.7***	92%	+30.2**	65%
	(2.2)		(2.7)		(2.1)	
Mid-month days (Mid)	+45.3**	47%	+30.3*	67%	+15.7	35%
	(2.1)		(1.9)		(1.1)	
Panel B: Excluding the 2001–02, 2009 and 2020 crash episodes						
All other days (Rest)	+50.5***	52%	+45.9***	91%	+35.0**	69%
	(2.6)		(3.1)		(2.5)	
Mid-month days (Mid)	+46.1**	48%	+30.6**	66%	+15.9	34%
	(2.2)		(2.0)		(1.1)	

Notes: Column (1) reports the average abnormal WML return during the PreTOM window, in basis points per month. The benchmark is the average daily WML return outside the PreTOM window, either on Rest or Mid days. Column (2) expresses the abnormal PreTOM return as a percentage of the average monthly WML return over the full sample. Column (3) measures the contribution of the loser leg to the abnormal PreTOM WML return. Specifically, it captures the additional underperformance of loser stocks relative to the market during PreTOM compared with the benchmark days. Column (4) expresses this contribution as a share of the abnormal WML return reported in column (1). Columns (5) and (6) report the corresponding loser-leg contribution using a CAPM-based market adjustment with a trailing one-year market beta, re-estimated each day from the previous 252 trading days and averaging 1.17. Column (5) reports the contribution in basis points, and column (6) expresses it as a share of the abnormal WML return. Columns (1) to (4) repeat Table 6. The sample covers 1980–2025 and uses value-weighted momentum portfolios with fixed monthly sorting. Rest comprises all days outside the PreTOM window. Mid comprises the middle of the same calendar month, from the fourth trading day through the day before the PreTOM window begins. Panel B repeats the table excluding the momentum-crash episodes August 2001 to March 2002, January to September 2009, and February to May 2020. t-statistics in parentheses are from paired monthly contrasts: the window-minus-benchmark difference is computed month by month and its mean is tested against zero across months. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Bootstrap intervals, alternative benchmarks, and subperiod analyses are reported in Internet Appendix Table IA.26.

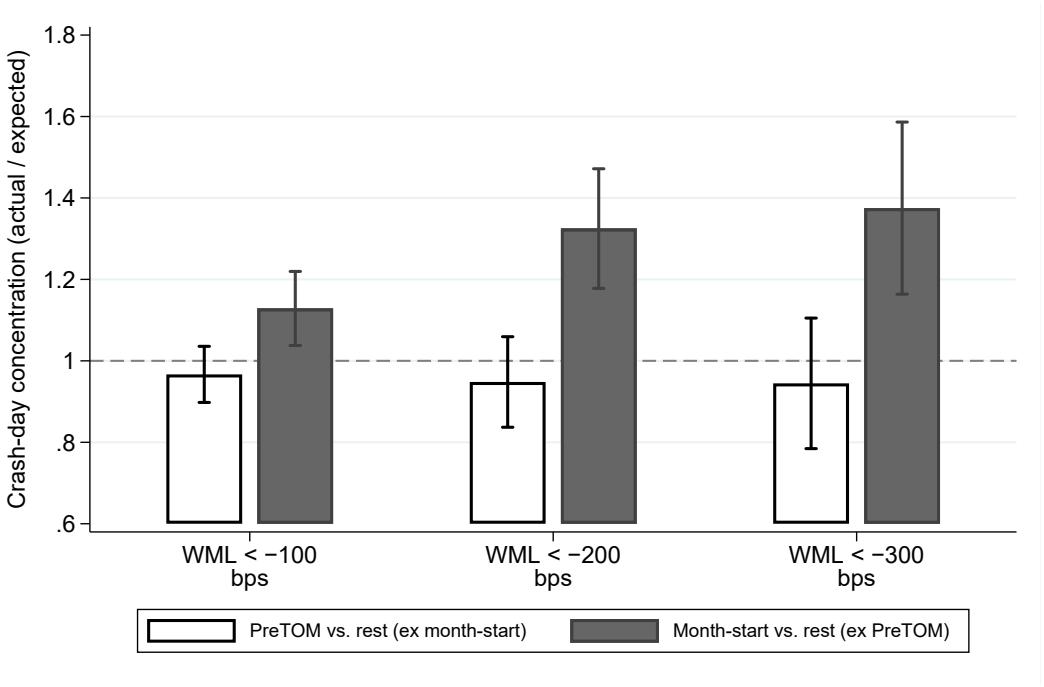


Figure IA.7: Crash-day concentration by calendar window. Each bar plots the ratio of actual crash-day share to calendar share; whiskers show 95% confidence intervals. The dashed horizontal line marks a ratio of one, the value expected if crash days were distributed uniformly across the calendar. Daily WML returns are value-weighted, 1980–2025. Deciles use NYSE breakpoints and are held fixed within each calendar month.

Table IA.13: Mutual-Fund Quantity Margins by Flow Bin, 1991–2024

	Inflow	Outflow quintile				
	intervals	Q1	Q2	Q3	Q4	Q5
	0.0%	0.7%	2.1%	3.6%	6.1%	17.5%
Quantity margin: liquidation beyond pro rata						
D1 (losers)	5.21	4.47	5.46	6.87	7.31	4.27
D2–D3	2.57	2.10	2.79	3.95	3.99	1.28
D4–D7	2.54	1.71	2.22	3.23	3.36	1.53
D8–D9	3.76	2.04	2.51	3.47	3.88	2.41
D10 (winners)	5.64	2.67	3.20	4.10	4.79	3.40
D1 − D4–D7	2.67	2.76	3.24	3.64	3.94	2.74
Within-fund-report estimates						
Loser, at zero outflow: 4.45 percentage points ($t = 9.28$)						
Loser × Outflow: 0.0312 ($t = 1.59$)						
0.19 percentage points per one-s.d. outflow, 95% CI [−0.04, 0.42]						

Notes: The quantity margin of Table 10, across bins of the fund's net flow. The table reports share reduction relative to the flow-adjusted benchmark for all positions, closures included. Outflow quintiles are formed across fund-reports in net outflow, Q1 the smallest, and the percentage beneath each quintile label is that column's mean net outflow. Sample, fixed effects, clustering and the delisting screen are as in Table 10. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.14: Loser Position Closures by Distance from the 52-Week High

Panel A: Loser positions by quintile of distance below the 52-week high

	Q1 (nearest)	Q2	Q3	Q4	Q5 (farthest)
Mean distance below the high	0.25	0.38	0.47	0.56	0.71
Share of positions closed (%)	10.1	11.5	13.1	15.1	21.5
Liquidation beyond pro rata (pp)	1.3	2.9	4.0	6.3	12.4
Positions	745,276	745,428	745,461	745,428	745,615

Panel B: Within fund-report regressions

	Losers (D1)				Middle (D2–D9)	
	Closure (1)	Closure (2)	Closure (3)	Excess liq. (4)	Closure (5)	Closure (6)
Distance from high (1 s.d.)	4.85***	4.73***	2.89***	4.71***	2.36***	1.48***
	(15.11)	(13.81)	(11.79)	(11.97)	(25.63)	(18.79)
Distance × Outflow		4.50***	1.82			2.47***
		(3.16)	(1.55)			(4.50)
Distance × Inflow		−0.01	0.39			0.26
		(−0.01)	(0.54)			(1.22)
Position controls	No	No	Yes	No	No	Yes
Fund × report FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent variable (%)	14.2	14.2	14.2	5.4	9.4	9.4
Positions	3,727,208	3,727,208	3,727,208	3,727,208	39,874,446	39,874,446
Fund-reports	254,267	254,267	254,267	254,267	365,858	365,858

Notes: Mutual-fund positions in momentum losers (bottom decile, Panel A and columns 1–4) and in the middle deciles (columns 5–6), 1991–2024, with the sample, closure indicator and liquidation beyond pro rata of Table 10. Distance from the high is one minus the ratio of the stock's price to its trailing 252-day maximum (Jensen et al., 2023), measured at the report date that opens the interval. Panel A sorts loser positions into quintiles of that distance at each report date and reports raw means. Panel B regresses the closure indicator or liquidation beyond pro rata on distance, standardized within report date among the positions of each sample, its interactions with the fund's outflow and inflow magnitudes, and, in columns 3 and 6, the position controls of Table 11 and their outflow interactions. Coefficients are in percentage points. t-statistics in parentheses, standard errors three-way clustered by fund, by stock and by report date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.15: Distance from the 52-Week High, Window Specificity, and the Winner and Fifth Deciles

	Q1 nearest	Q2	Q3	Q4	Q5 farthest	Q5 − Q1
Does either relationship steepen inside the window? (bottom decile)						
Selling, distance ×						0.00019
PreTOM, firm and date FE						(0.35)
Returns, PreTOM − Rest,						−7.29*
monthly Q5 − Q1 spreads						(−1.81)
Top decile (winners)						
Mean price / 52-week high	0.98	0.94	0.88	0.80	0.61	
PreTOM, bps per day	−0.67	−2.02	−0.83	−2.61	−5.47	−4.80
						(−1.56)
Rest, bps per day	+6.77	+8.10	+9.54	+8.67	+6.80	+0.04
						(0.02)
PreTOM − Rest						−5.00
						(−1.39)
Fifth decile						
Mean price / 52-week high	0.95	0.89	0.85	0.78	0.63	
PreTOM, bps per day	−3.80	−4.06	−5.18	−2.79	−4.63	−0.83
						(−0.34)
Rest, bps per day	+4.81	+6.97	+6.93	+7.18	+6.85	+2.04
						(1.32)
PreTOM − Rest						−3.04
						(−1.11)

Notes: The sort of Table 12 applied to the top and fifth momentum deciles. Within each decile, stocks are sorted at the prior month-end by split-adjusted price divided by the trailing 252-day maximum (Jensen et al., 2023); Q1 is nearest the high, Q5 farthest. Entries are value-weighted returns above the risk-free rate, in basis points per day, 1980–2025. PreTOM is $[\tau-9, \tau-4]$ and Rest comprises the other days. Quintile means and Q5−Q1 spreads average daily observations; PreTOM−Rest averages monthly spread differences and uses Newey–West standard errors. The winner point estimate has the same sign but is statistically indistinguishable from zero, and the fifth decile shows no window spread. The pronounced spread is concentrated in the loser leg. We report these without testing whether the legs differ. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.16: The Window Return and Subsequent Returns under Alternative Outcome Windows

Recovery accounting, U.S. losers, alternative outcome windows				
Outcome window	PreTOM (β_1)	Placebo (β_2)	$\beta_1 - \beta_2$	N
Omitted days $[\tau{-}3, \tau]$	-0.085^{***}	-0.011^{***}	-0.074^{***}	476,736
	(-17.8)	(-3.8)	(-15.4)	
Next three days $[\tau{+}1, \tau{+}3]$	-0.012^{***}	$+0.003$	-0.015^{***}	476,736
	(-3.6)	(1.1)	(-4.2)	
Both, $[\tau{-}3, \tau{+}3]$	-0.097^{***}	-0.008^{*}	-0.089^{***}	476,736
	(-15.9)	(-1.9)	(-14.8)	

Notes: Equation (IA.6), using the alternative cumulative market-adjusted outcome windows shown and the delisting treatment of Table IA.6, Panel A. Regressors are the PreTOM and Early returns, with cohort fixed effects. Parentheses report t-statistics; standard errors are two-way clustered by stock and outcome-ending calendar month. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.16 Beta Controls

The Loser×PreTOM effect could in principle be a mechanical transmission of the market's own PreTOM underperformance through higher loser betas rather than targeted selling of losers. To check this we add the 60-month rolling CAPM beta of Jensen et al. (2023), centered at the cross-sectional average beta on each date, and saturate the baseline with β, PreTOM$\times\beta$, Loser$\times\beta$ and Loser×PreTOM$\times\beta$. Table IA.17 reports the two coefficients that bear on the question, for value-weighted returns, equal-weighted returns and TAQ net selling pressure. The window coefficient survives the controls at close to its original size: value-weighted it moves from -7.30 to -7.02 basis points per day, and in selling pressure from $+0.0017$ to $+0.0018$. The triple interaction, which is the additional loser-specific change in the beta slope during the window, is indistinguishable from zero in all three outcomes. The equal-weighted specification is the one place where the window coefficient moves materially, from -3.35 to -2.33, so part of the equal-weighted effect is beta-related even though the value-weighted and selling-pressure results are not. PreTOM$\times\beta$ absorbs any window move transmitted through a beta common to all stocks, and Loser$\times\beta$ the level, so the window coefficient measures loser underperformance beyond beta.

Table IA.17: Beta Controls in Value-Weighted Returns, Equal-Weighted Returns, and TAQ Selling Pressure

	Returns, VW (bps/day)		Returns, EW (bps/day)		TAQ net selling (% of volume)	
	Baseline	With beta	Baseline	With beta	Baseline	With beta
	(1)	(2)	(3)	(4)	(5)	(6)
Loser × PreTOM	−7.303***	−7.024***	−3.350**	−2.331*	0.17**	0.18**
	(−3.09)	(−3.55)	(−2.44)	(−1.85)	(2.05)	(2.17)
Loser × PreTOM × β		3.050		0.203		0.03
		(1.36)		(0.16)		(0.48)
Observations	44,638,274	44,638,263	44,638,274	44,638,263	15,972,786	15,972,786

Notes: Each pair of columns is a separate outcome. Columns (1)–(4) are daily market-adjusted stock returns in basis points, CRSP 1980–2025, weighted by lagged capitalization (VW) and equally across stocks (EW). Columns (5)–(6) are CRSP/TAQ (sell volume − buy volume)/total volume, 2003–2022. The Baseline columns estimate equation (1) on the subsample with a nonmissing beta; the With beta columns add β, PreTOM × β, Loser × β and Loser × PreTOM × β to that specification. The two specifications are estimated on samples that differ by a small number of observations, so both counts are reported. Beta is the 60-month rolling CAPM beta of Jensen et al. (2023), demeaned within date using the weights of the corresponding column. All specifications include stock and date fixed effects. Parentheses report t-statistics with standard errors two-way clustered by stock and date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.17 European T+3-to-T+2 Settlement Transition

This section reports country-level evidence from the synchronized European migration from T+3 to T+2 settlement on October 6, 2014, implemented under the EU Central Securities Depositories Regulation (CSDR). Thirteen markets migrated their equity settlement on that date, namely Austria, Belgium, Denmark, Finland, France, Ireland, Italy, the Netherlands, Norway, Portugal, Sweden, Switzerland and the United Kingdom. Germany did not: its equities already settled T+2, and the German instruments that migrated in October 2014 were a category of debt. Spain did not either; Spanish equities migrated with effect from trades on 29 September 2016. Both markets stay in the international panel, neither is treated here, and Spain is available as a comparison market in a window that ends before its own migration. The settlement mechanism predicts that this reform should move loser-selling pressure from $\tau-5$ to $\tau-4$. We estimate this boundary-day contrast separately by country and aggregate the estimates across countries.

Before turning to the difference-in-differences estimates, Figure IA.8 establishes that the transition markets share the U.S. loser-selling window. Pooled across the thirteen markets, mean loser returns in excess of each country's own value-weighted market, the benchmark used in the difference-in-differences below, decline from mid-month into a trough just before month-end and rise at month-end.[5] Over the full sample (Panel A) the trough region runs from $[\tau-9, \tau-4]$, at -5.31 basis points per day ($t = -3.91$), through $[\tau-8, \tau-3]$ at -5.87 ($t = -4.51$) and $[\tau-7, \tau-2]$ at -6.16 ($t = -4.77$); these windows are statistically indistinguishable from one another and match the location of the U.S. trough (Table IA.20). In the pre-reform T+3 period (Panel B) the trough sits in the same region, at -6.11 basis points per day ($t = -3.44$) for $[\tau-9, \tau-4]$, -5.94 ($t = -3.45$) for $[\tau-8, \tau-3]$ and -6.56 ($t = -3.87$) for $[\tau-7, \tau-2]$, so the window that anchors the boundary-day comparison is identified by data that predate the reform. Both panels place the trough in the same region. The one-day shift itself is identified by the boundary difference-in-differences below, not by comparing trough locations across eras.

For each country c, we estimate equation (6) from the main text. The dependent variable is the value-weighted D1 loser portfolio return in excess of the country market return, mea-

[5] The location of the trough is insensitive to the benchmark: subtracting Kenneth French's regional market return instead, or the U.S. one-month T-bill rate, or using raw local-currency returns, places it in the same region (untabulated).

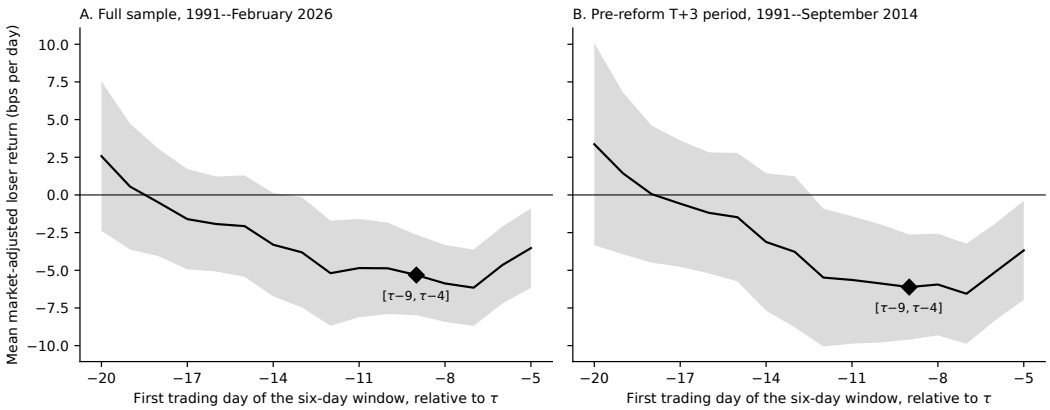


Figure IA.8: Sliding six-day window analysis for the European transition markets. Each point plots the mean daily value-weighted loser return in excess of that country's own value-weighted market return for a six-day window starting at the indicated trading day, pooled across the thirteen European T+3-to-T+2 transition markets (equal-weighted within month; country-months with all six days present). Panel A uses the full sample (1991–February 2026); Panel B uses the pre-reform T+3 period (1991–September 2014). The diamond marks the PreTOM window $[\tau-9, \tau-4]$. Shaded areas are 95% confidence intervals from Newey-West standard errors (5 lags) on monthly means.

sured in basis points. The treated days are $\tau-5$ and $\tau-4$. The omitted reference category is a mid-month control window $[\tau-15, \tau-10]$. The country-level contrast is $\kappa_c = \theta_{c,\tau-5} - \theta_{c,\tau-4}$, where a positive value indicates that the old boundary day $\tau-5$ improves relative to the new boundary day $\tau-4$ after the reform.

Figure IA.9 reports the country-level estimates for the markets that transitioned. Ten of the thirteen country-level contrasts are positive and the inverse-variance-weighted pooled estimate is +14.73, while every individual country's confidence interval is wide: the sign concordance across countries is informative about the breadth of the response. Because the reform was synchronized, we treat it as one European event observed across thirteen markets rather than as thirteen independent experiments. The three negative estimates are Portugal (−21.86), Switzerland (−3.70) and Belgium (−1.99), none of them more than 1.0 standard errors below zero. Switzerland is the only market outside the EU and the CSDR scope, and excluding it raises the pooled estimate to +17.93. The largest single estimate, Ireland at +89.80, also carries the widest confidence interval and the shortest sample.

The pooled estimate remains positive after dropping any single country, ranging from

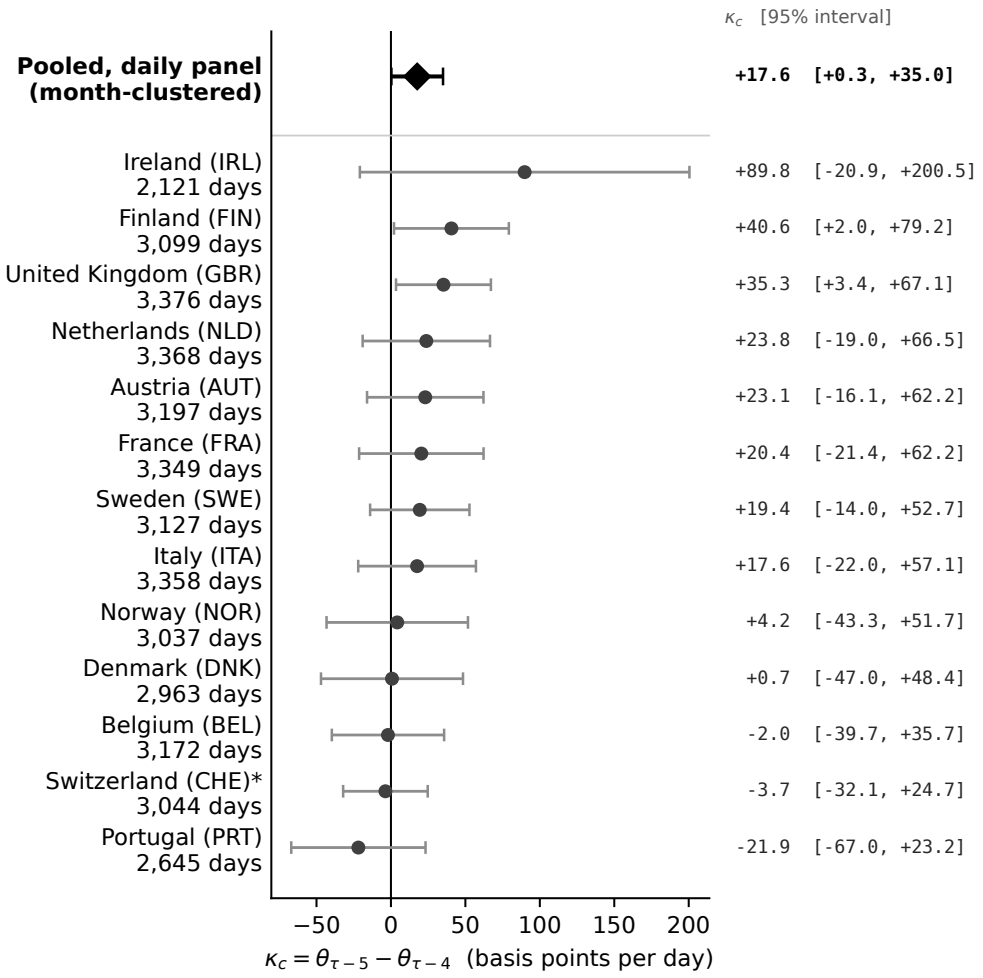


Figure IA.9: Country-level boundary-day contrasts for the European T+3-to-T+2 transition. Each dot is one market's difference-in-differences contrast $\kappa_c = \theta_{c,\tau-5} - \theta_{c,\tau-4}$ from equation (6), the improvement of the old boundary day $\tau-5$ relative to the new boundary day $\tau-4$ after the reform of October 6, 2014. The dependent variable is the value-weighted D1 loser return in excess of that country's own market return, in basis points per day, and the omitted reference category is the mid-month window $[\tau-15, \tau-10]$. Country whiskers are 95% confidence intervals from date-clustered standard errors, and each market's estimate, interval and trading-day count are printed at the right. The diamond is not an average of the country estimates: it is the pooled daily-panel contrast with country and calendar-month fixed effects and common slopes, clustered by calendar month, which is the specification the main text reports and the one that lets the thirteen markets move together. Its whisker is that estimate's own 95% interval, $[+0.31, +34.99]$. Sample: 1991–February 2026. *Switzerland is the one transition market outside the European Union and the scope of the CSDR.

+11.98 to +17.93 basis points per day. Inference has to allow the thirteen markets to move together. They changed settlement on the same date and share most of their month-to-month variation, so the standard error implied by the inverse-variance weights, $(\sum_c SE_c^{-2})^{-1/2} = 5.61$, is the standard error of an average of independent estimates and is too small. We therefore estimate the contrast on the pooled daily panel of the thirteen markets, with country and calendar-month fixed effects and common slopes, clustering by calendar month so that the markets may move together within a month. This is the specification the main text reports. The loser contrast is +17.65 basis points per day with a standard error of 8.82 ($t = 2.00$, $p = 0.046$ on 39,856 country-days in 421 month clusters), and the corresponding 95% interval is $[+0.31, +34.99]$. The benchmark-free WML contrast is −25.16 with a standard error of 13.60 ($t = -1.85$, $p = 0.065$); it matters because a long-short difference cannot inherit a benchmark choice. The alternative covariances agree on sign and differ on width. Clustering by calendar date gives $t = 2.08$ for the loser contrast and $t = -1.93$ for WML. Two-way clustering by country and calendar month, which leaves twelve degrees of freedom, gives $p = 0.067$ and $p = 0.063$. Driscoll–Kraay standard errors on the monthly series with a Bartlett lag of five give $t = 2.19$ and $t = -1.89$. The European transition therefore moves the boundary in the predicted direction on both outcomes under every scheme we report, with a magnitude that the data do not pin down.

IA.18 U.S. T+3-to-T+2 Settlement Transition

An earlier U.S. settlement reform, the September 5, 2017 transition from T+3 to T+2 equity settlement, provides a directional check on the same comparative static. The dash-for-cash mechanism predicts a one-day shift in loser-selling pressure at this reform, from $\tau-5$ (the T+3-era boundary day) to $\tau-4$ (the T+2-era boundary day).

Table IA.18 reports the boundary-day DiD using the full T+3 regime as the pre-period (June 1995 – August 2017) and the T+2 era as the post-period (October 2017 – April 2024, excluding both September 2017, the regime-boundary month, and the T+1 era from May 2024 onward). The $\tau-5$ vs. $\tau-4$ DiD is +13.08 basis points ($F = 0.27$, $p = 0.60$): the point estimate is in the predicted direction (selling pressure moves toward $\tau-4$) but is not statistically distinguishable from zero.

Table IA.18: T+3→T+2 Settlement Boundary-Day Difference-in-Differences

Panel A: 2×2 Cell Means (loser market-adjusted return, bps/day)			
	Pre-T+2 (Jun 1995–Aug 2017)	Post-T+2 (Oct 2017–Apr 2024)	Diff
$\tau-5$	−6.09	−11.26	−5.16
	($N = 267$)	($N = 79$)	
$\tau-4$	−6.86	−25.10	−18.24
	($N = 267$)	($N = 79$)	
Diff ($\tau-5$ minus $\tau-4$)	0.77	13.85	13.08

Panel B: DiD Regression with Early-Month Control	
	Coefficient
$\theta_{\tau-5}$: $\mathbf{1}_{[\tau-5]} \times$ Post T+2	−7.92
	(21.32)
$\theta_{\tau-4}$: $\mathbf{1}_{[\tau-4]} \times$ Post T+2	−21.00
	(21.73)
$\theta_{\tau-5} - \theta_{\tau-4}$ (boundary shift)	13.08
	(25.22)
	$F = 0.27$, $p = 0.604$

Notes: Value-weighted market-adjusted loser returns in basis points around the September 5, 2017 settlement reform. Treated days are $\tau-5$ and $\tau-4$; controls are $[\tau'+5, \tau'+8]$, trading days 5–8 of the month. The pre-period is June 1995–August 2017 and the post-period October 2017–April 2024. September 2017 and the subsequent T+1 era are excluded. Parentheses report heteroskedasticity-robust standard errors. The boundary contrast is $\theta_{\tau-5} - \theta_{\tau-4}$; the common early-month control cancels from this difference.

IA.19 S&P 500 Membership Robustness

A natural concern is that the PreTOM effect is concentrated among small-cap or thinly traded stocks. The effect is also present among S&P 500 constituents. The Loser × PreTOM coefficient is -8.77 ($t = -2.71$) among S&P 500 stocks and -4.62 ($t = -2.53$) among non-constituents. The effect is not confined to small or illiquid stocks. The same split makes an S&P 500 index-rebalancing story an unlikely driver. Index funds must trade constituents, so pressure generated by index rebalancing would concentrate in S&P 500 stocks, and the effect is present among constituents and non-constituents alike. Rebalancing toward fixed weights also carries the wrong sign for this mechanism. A rebalancer buys the assets that have fallen and sells those that have risen, so it buys momentum losers and sells momentum winners, whereas the window effect is losers being sold.

Table IA.19: PreTOM Effect by S&P 500 Membership

	Non-S&P 500 (1)	S&P 500 (2)
Loser	1.806*	4.018**
	(1.84)	(2.23)
Loser × PreTOM	−4.622**	−8.769***
	(−2.53)	(−2.71)
Fixed effects	Stock, Date	Stock, Date
Observations	47,863,849	5,460,280
Within R^2	< 0.0001	< 0.0001

Notes: Daily market-adjusted stock returns in basis points, CRSP 1980–2025, weighted by lagged capitalization. The two columns estimate the specification separately on S&P 500 nonconstituents and constituents, using each stock's membership on the date of the observation. Sixteen stock-days with a missing membership flag are excluded. Both columns include stock and date fixed effects. Parentheses report t-statistics with standard errors two-way clustered by stock and date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.20 The Window

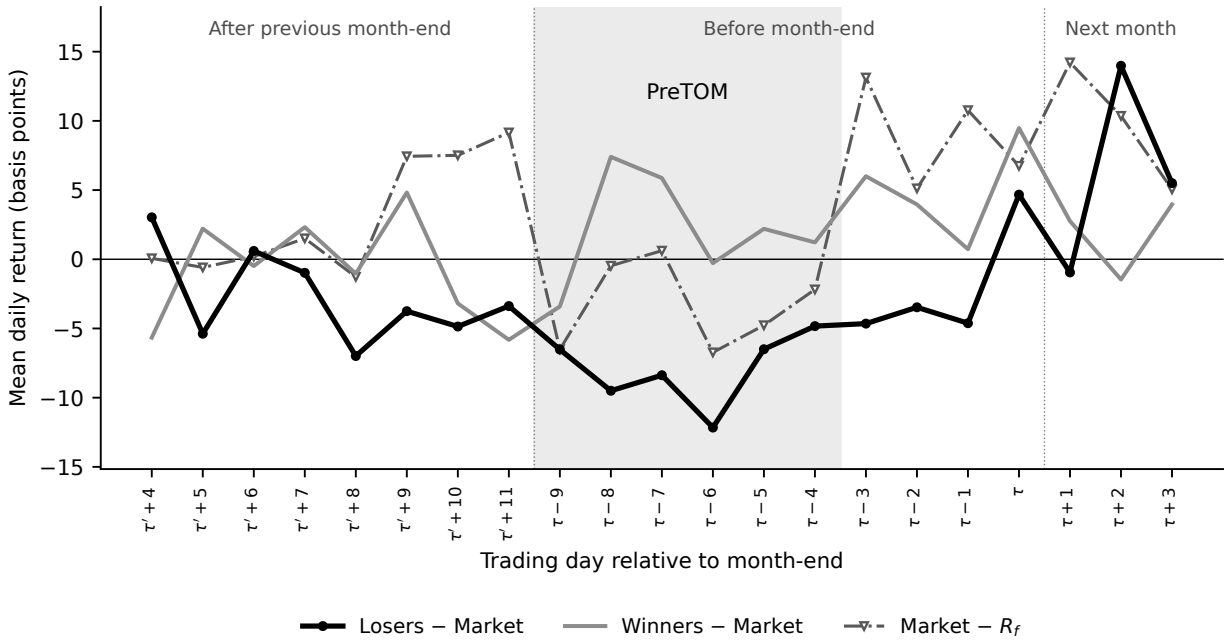


Figure IA.10: Daily return profile by trading day within the month. Mean daily value-weighted returns, 1980–2025, in basis points: the bottom momentum decile relative to the market, the top momentum decile relative to the market, and the market excess return. Each day is assigned to the nearer month-end. Positions $\tau'+4$ through $\tau'+11$ are counted forward from the previous month-end τ', positions $\tau-9$ through τ backward from the current month-end τ, and $\tau+1$ through $\tau+3$ are the first three trading days of the next month. Because deciles are re-sorted at each month-end, the mean on $\tau+k$ equals the mean on $\tau'+k$; the axis places these days after τ to show the turn of the month in sequence. The shaded region marks the PreTOM window $[\tau-9, \tau-4]$. Individual-day standard errors are approximately 4 to 6 basis points.

Table IA.20 reports the sensitivity of the window return to the window's boundaries. Panel A shows a plateau: the WML window rate is 10.2 basis points per day for $[\tau-7, \tau-4]$, 11.6 for $[\tau-8, \tau-4]$, 10.2 for $[\tau-9, \tau-4]$, and 8.8 for $[\tau-10, \tau-4]$, and the loser differential is significant from $k = 7$ on. Panel B shows that the six-day blocks from $[\tau-7, \tau-2]$ to $[\tau-11, \tau-6]$ all earn 9 to 11 basis points per day and that blocks starting earlier than $\tau-12$ earn nothing; the block that fits best in the full sample is $[\tau-8, \tau-3]$ at 11.4, and Etula et al.'s $[\tau-8, \tau-4]$ earns 11.6. Panel C reports performance in 2001–2025 for the paper's

window and for two windows selected using 1980–2000. The paper's window earns 9.06 basis points per day in the later sample ($t = 2.02$ on the difference from other days). It was not selected by that training-period search. The search-selected windows, $[\tau{-}8, \tau{-}4]$ and $[\tau{-}11, \tau{-}6]$, earn 10.45 and 6.07 basis points per day, respectively, in 2001–2025.

Table IA.20: Window Sensitivity and Out-of-Sample Evaluation

Panel A: Windows $[\tau{-}k, \tau{-}4]$*,* $k = 4, \ldots, 12$*, 1980–2025*								
		WML (bps/day)			Losers − Mkt (bps/day)			\$1 in window
Window	Days	Window	Other days	t(diff)	Window	Other days	t(diff)	
$[\tau{-}4, \tau{-}4]$	1	6.05	4.56	0.22	−4.83	−2.82	−0.45	1.31
$[\tau{-}5, \tau{-}4]$	2	7.38	4.32	0.60	−5.67	−2.61	−0.86	1.99
$[\tau{-}6, \tau{-}4]$	3	8.88	3.90	1.20	−7.83	−2.06	−1.83	3.59
$[\tau{-}7, \tau{-}4]$	4	10.22	3.28	2.01	−7.97	−1.67	−2.38	7.33
$[\tau{-}8, \tau{-}4]$	5	11.56	2.40	2.48	−8.27	−1.18	−2.44	16.97
$[\tau{-}9, \tau{-}4]$ (paper)	6	10.15	2.33	2.41	−7.98	−0.79	−2.99	18.78
$[\tau{-}10, \tau{-}4]$	7	8.78	2.51	2.08	−7.41	−0.57	−3.08	18.20
$[\tau{-}11, \tau{-}4]$	8	9.38	1.65	2.39	−8.13	0.42	−3.50	35.97
$[\tau{-}12, \tau{-}4]$	9	8.49	1.61	2.11	−7.45	0.67	−3.28	35.98

Table IA.20: Window Sensitivity and Out-of-Sample Evaluation (continued)

		WML (bps/day)			Losers − Mkt (bps/day)			\$1 in window
Window	Days	Window	Other days	t(diff)	Window	Other days	t(diff)	
Panel B: Every six-day block $[\tau-(s+5), \tau-s]$*, 1980–2025*								
$[\tau-5, \tau]$	6	7.17	3.56	1.14	−3.23	−2.76	−0.20	7.52
$[\tau-6, \tau-1]$	6	8.34	3.11	1.69	−6.04	−1.64	−2.06	10.92
$[\tau-7, \tau-2]$	6	9.83	2.52	2.63	−6.66	−1.38	−2.46	17.80
$[\tau-8, \tau-3]$	6	11.41	1.86	2.78	−7.67	−0.97	−2.44	28.80
$[\tau-9, \tau-4]$ (paper)	6	10.15	2.33	2.41	−7.98	−0.79	−2.99	18.78
$[\tau-10, \tau-5]$	6	9.24	2.73	2.08	−7.84	−0.85	−2.83	13.84
$[\tau-11, \tau-6]$	6	10.05	2.43	2.12	−8.95	−0.43	−3.10	18.08
$[\tau-12, \tau-7]$	6	8.30	3.09	1.50	−7.26	−1.11	−2.24	10.01
$[\tau-13, \tau-8]$	6	5.93	4.08	0.56	−5.53	−1.83	−1.54	4.49
$[\tau-14, \tau-9]$	6	3.38	5.13	−0.45	−3.68	−2.57	−0.37	1.89
$[\tau-15, \tau-10]$	6	2.95	5.32	−0.65	−3.01	−2.89	−0.04	1.60

Table IA.20: Window Sensitivity and Out-of-Sample Evaluation (continued)

Panel C: Performance in 2001–2025: training-selected windows and the paper's window

Window	Sample	Window	Other days	t(diff)	Window	Other days	t(diff)	
Paper window $[\tau-9, \tau-4]$	1980–2000	11.45	6.83	1.39	−10.67	−4.01	−2.92	5.19
Paper window $[\tau-9, \tau-4]$	2001–2025	9.06	−1.44	2.02	−5.72	1.92	−1.95	3.62
Best $[\tau-k, \tau-4]$ in 1980–2000: $[\tau-8, \tau-4]$	1980–2000	12.88	6.70	1.78	−10.66	−4.47	−2.51	4.72
Best $[\tau-k, \tau-4]$ in 1980–2000: $[\tau-8, \tau-4]$	2001–2025	10.45	−1.22	1.93	−6.27	1.59	−1.62	3.59
Best six-day block in 1980–2000: $[\tau-11, \tau-6]$	1980–2000	14.80	5.75	2.60	−13.23	−3.14	−4.76	8.60
Best six-day block in 1980–2000: $[\tau-11, \tau-6]$	2001–2025	6.07	−0.37	1.09	−5.36	1.85	−1.53	2.10
Etula et al. (2020) window $[\tau-8, \tau-4]$	1980–2000	12.88	6.70	1.78	−10.66	−4.47	−2.51	4.72
Etula et al. (2020) window $[\tau-8, \tau-4]$	2001–2025	10.45	−1.22	1.93	−6.27	1.59	−1.62	3.59

Notes: Daily value-weighted returns with fixed monthly sorting, in basis points. Window is the mean daily return on the window's days. Other days is the mean, across months, of each month's average daily return on the remaining days, so for the paper's window it differs slightly from the pooled daily means of Table 1. t(diff) is the Newey–West t-statistic, with three lags, on the monthly difference between the window and the other days. "$1 in window" compounds WML only on window days over the stated sample. Panel C evaluates in 2001–2025 the highest-return window and six-day block selected using 1980–2000. It also reports the paper and Etula et al. (2020) windows; the paper's window was not selected by that search.

IA.21 Supporting Tables and Figures for the Main Text

Table IA.21 reports the crash concentration with outliers dropped; Table IA.22 the bid-ask spread interaction; Table IA.23 the subperiod estimates of equation (1); Table IA.24 additional specifications of the within-cohort regression; Table IA.25 inference and weighting checks for its international version; and Table IA.26 the attribution in levels, over 1927–2025, and by subperiod.

Table IA.21: Robustness of Crash-Day Shares by Calendar Window

Sample	Threshold (bps)	Crash days	Month-start share	Calendar share	Ratio	z
All days, 1980–2025	−100	1376	0.301	0.267	1.13	2.88
	−200	572	0.353	0.267	1.32	4.68
	−300	281	0.367	0.267	1.38	3.79
Excluding the 2001–02, 2009, 2020 crash episodes	−100	1270	0.298	0.266	1.12	2.53
	−200	504	0.347	0.266	1.30	4.11
	−300	241	0.357	0.266	1.34	3.18
Dropping the 20 most extreme WML days	−100	1366	0.302	0.267	1.13	2.92
	−200	562	0.356	0.267	1.33	4.78
	−300	271	0.373	0.267	1.40	3.95
Dropping the 1% most extreme WML days	−100	1321	0.298	0.266	1.12	2.63
	−200	517	0.352	0.266	1.32	4.41
	−300	226	0.367	0.266	1.38	3.44
WML residual on trailing 126-day conditional beta	−100	1224	0.321	0.267	1.20	4.31
	−200	509	0.352	0.267	1.32	4.34
	−300	222	0.365	0.267	1.37	3.31

Notes: A crash day has value-weighted WML below the stated threshold. Month-start is $[\tau+1, \tau+4]$; the comparison universe excludes PreTOM. The z-statistic compares the month-start share of crashes with its share of eligible trading days, using an independent-day binomial variance. Dependence-adjusted inference is discussed in Section IA.5. The final block uses WML residualized on a trailing 126-day market beta.

Table IA.22: PreTOM Effect on Losers and the Bid-Ask Spread

	(1) VW	(2) EW
PreTOM (β)	-9.039^{***}	-4.237^{*}
	(-2.92)	(-1.96)
$\text{BAS}_{\text{prior}}$ (γ_1)	99.510^{***}	247.814^{***}
	(8.08)	(24.68)
PreTOM $\times$ $\text{BAS}_{\text{prior}}$ (γ_2)	39.040	29.537^{***}
	(1.41)	(2.82)
Sample	1980–2025	1980–2025
Fixed effects	Stock, Month	Stock, Month
Observations	9,069,541	9,069,541

Notes: Sample: bottom-decile (D1) momentum stocks, 1980–2025. Dependent variable is $\text{ExRet}_{i,t} = r_{i,t} - r_t^m$ (market-adjusted return) in basis points, as defined in equation (1). PreTOM is an indicator for the six trading days $[\tau-9, \tau-4]$ before month-end (τ is the last trading day). $\text{BAS}_{\text{prior}}$ is the mean bid-ask spread of stock i over calendar month $m-1$, as a fraction of the quote midpoint, computed from all CRSP stocks regardless of decile membership in month $m-1$; stock-months with fewer than 10 valid prior-month BAS days are excluded. The estimation sample is the full D1 panel with a valid $\text{BAS}_{\text{prior}}$; no restriction is placed on the availability of the current-day spread. All specifications include stock and year-month fixed effects. t-statistics are in parentheses; standard errors are two-way clustered by stock and date. Column 1 weights observations by lagged market capitalization. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.23: The PreTOM Effect by Subperiod

	(1) 1980–2001	(2) 2002–2025
Loser	0.708	5.648***
	(0.55)	(2.73)
Loser × PreTOM	−5.146**	−9.018**
	(−2.07)	(−2.35)
Fixed effects	Stock, Date	Stock, Date
Observations	29,936,488	23,387,655
Within R^2	<0.0001	<0.0001

Notes: Estimates of equation (1) on subperiods. Dependent variable is the daily market-adjusted return ($r_{i,t} - r_t^{mkt}$, in excess of the daily value-weighted market return from Kenneth French's data library), in basis points. Returns are value-weighted. Sample split at the end of 2001, as in Table 5. All specifications include stock and date fixed effects. t-statistics are in parentheses; standard errors are two-way clustered by stock and date. Momentum deciles use NYSE breakpoints applied to all NYSE, AMEX, and NASDAQ stocks in CRSP; assignments are held constant within each calendar month. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.24: The Window Return and Subsequent Returns, Additional Specifications

Panel A: Survival required versus delisting returns, U.S. losers, 1980–2025

	Survival required			With delisting returns		
Outcome horizon	β_1	$\beta_1 - \beta_2$	N	β_1	$\beta_1 - \beta_2$	N
$[\tau{+}1, \tau{+}3]$	-0.016^{***}	-0.016^{***}	474,986	-0.012^{***}	-0.015^{***}	476,736
	(-5.2)	(-5.2)		(-3.6)	(-4.2)	
$[\tau{+}1, \tau{+}21]$	-0.036^{***}	-0.032^{***}	469,464	-0.010	-0.028^{***}	476,736
	(-4.9)	(-4.5)		(-1.3)	(-3.6)	
Months 2 to 3	$+0.027^{***}$	$+0.004$	454,751	$+0.059^{***}$	$+0.004$	469,800
	(3.0)	(0.5)		(6.2)	(0.4)	
Months 4 to 6	$+0.056^{***}$	$+0.019^{*}$	433,075			
	(5.1)	(1.8)				
Months 7 to 12	$+0.082^{***}$	-0.003	393,758	$+0.145^{***}$	$+0.012$	433,433
	(5.6)	(-0.2)		(9.7)	(0.8)	
Months 13 to 18	$+0.051^{***}$	$+0.028^{*}$	361,178			
	(3.5)	(1.9)				
Months 19 to 36	$+0.073^{***}$	$+0.028$	283,924	$+0.160^{***}$	$+0.042$	361,378
	(2.7)	(1.0)		(5.4)	(1.4)	
Months 37 to 48	$+0.062^{***}$	$+0.046^{**}$	244,016			
	(3.3)	(2.1)				

Table IA.24: The Window Return and Subsequent Returns, Additional Specifications (continued)

Panel B: The month-end days $[\tau-3, \tau]$ as the regressor, and the era split

Sample	Regressor	Outcome	β_1	$\beta_1 - \beta_2$	N
1980–2025	$R^{[\tau-3,\tau]}$	$[\tau+2, \tau+4]$	-0.035^{***}	-0.036^{***}	474,544
			(-9.1)	(-8.2)	
1980–2025	$R^{[\tau-3,\tau]}$	$[\tau+2, \tau+22]$	-0.058^{***}	-0.054^{***}	468,958
			(-7.0)	(-6.9)	
2002–2025	$R^{[\tau-3,\tau]}$	$[\tau+2, \tau+4]$	-0.022^{***}	-0.024^{***}	197,027
			(-3.5)	(-3.5)	
2002–2025	$R^{[\tau-3,\tau]}$	$[\tau+2, \tau+22]$	-0.032^{**}	-0.038^{***}	195,023
			(-2.4)	(-3.0)	
2017:09–2025	$R^{[\tau-3,\tau]}$	$[\tau+2, \tau+4]$	-0.008	-0.016^{*}	69,612
			(-1.0)	(-1.8)	
1980–2001	R^{PreTOM}	$[\tau+1, \tau+3]$	-0.027^{***}		277,695
			(-7.0)		
2002–2025	R^{PreTOM}	$[\tau+1, \tau+3]$	-0.000		197,142
			(-0.1)		

Panel C: Midquote returns (bid-ask midpoints), 1993–2025

Sample	Regressor	Recovery over three days	Recovery over one month
1993–2025	$R^{[\tau-3,\tau]}$	3.4%	5.9%
2002–2025	$R^{[\tau-3,\tau]}$	1.5%	2.8%

Notes: Variants of equation (IA.6). Panel A compares samples requiring survival through the outcome horizon with samples appending delisting returns. Standard errors are two-way clustered by stock and outcome-ending calendar month; parentheses report t-statistics. Panel B uses returns over $[\tau-3, \tau]$ as the regressor and begins outcomes at $\tau+2$; it also reports regressions by era, with the sample split at the end of 2001 as in Table 5. Panel C uses bid-ask midpoint returns, 1993–2025. Recovery shares reverse the sign of the corresponding return slope. The treatment of overlapping horizons is discussed in Section IA.2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Horizons without an entry under delisting returns were not estimated on that basis.

Table IA.25: Inference and Weighting for International Non-Recovery

Sample	Horizon	β_1	β_2	$\beta_1 - \beta_2$	N
All 19 markets,	$[\tau{+}1, \tau{+}3]$	-0.017^{***}	-0.006	-0.011^{***}	442,048
two-way clusters		(-4.36)		(-2.68)	
All 19 markets,	$[\tau{+}1, \tau{+}21]$	-0.035^{***}	-0.005	-0.030^{***}	438,620
two-way clusters		(-4.61)		(-3.46)	
All 19 markets,	months 2–3	$+0.041^{***}$	$+0.044$	-0.003	430,750
two-way clusters		(3.15)		(-0.26)	
All 19 markets,	months 4–6	$+0.045^{***}$	$+0.052$	-0.008	418,744
two-way clusters		(3.99)		(-0.57)	
Europe, two-way clusters	$[\tau{+}1, \tau{+}3]$	-0.017^{***}	-0.003	-0.014^{**}	286,341
		(-3.15)		(-2.43)	
Europe, two-way clusters	$[\tau{+}1, \tau{+}21]$	-0.018	$+0.007$	-0.025^{**}	283,809
		(-1.59)		(-1.96)	
Europe, two-way clusters	months 2–3	$+0.069^{***}$	$+0.072$	-0.003	277,923
		(3.54)		(-0.14)	
Europe, two-way clusters	months 4–6	$+0.063^{***}$	$+0.069$	-0.006	269,169
		(3.76)		(-0.30)	
Asia-Pacific, two-way clusters	$[\tau{+}1, \tau{+}3]$	-0.017^{***}	-0.012	-0.005	155,707
		(-4.41)		(-0.97)	
Asia-Pacific, two-way clusters	$[\tau{+}1, \tau{+}21]$	-0.069^{***}	-0.030	-0.039^{***}	154,811
		(-9.68)		(-4.31)	
Asia-Pacific, two-way clusters	months 2–3	-0.012	-0.009	-0.003	152,827
		(-1.38)		(-0.24)	
Asia-Pacific, two-way clusters	months 4–6	$+0.012$	$+0.022$	-0.009	149,575
		(1.23)		(-0.76)	

Table IA.25: Inference and Weighting for International Non-Recovery (continued)

Sample	Horizon	β_1	β_2	$\beta_1 - \beta_2$	N
All 19 markets, equal country weights	$[\tau{+}1, \tau{+}3]$	-0.022^{***}	-0.008	-0.014^{**}	442,048
		(-4.22)		(-2.28)	
All 19 markets, equal country weights	$[\tau{+}1, \tau{+}21]$	-0.035^{***}	-0.018	-0.017	438,620
		(-3.19)		(-1.24)	
All 19 markets, equal country weights	months 2–3	$+0.053^{***}$	$+0.061$	-0.008	430,750
		(3.09)		(-0.43)	
All 19 markets, equal country weights	months 4–6	$+0.043^{***}$	$+0.074$	-0.032	418,744
		(3.03)		(-1.55)	
Asia-Pacific, equal country weights	$[\tau{+}1, \tau{+}3]$	-0.023^{***}	-0.013	-0.010	155,707
		(-3.01)		(-1.17)	
Asia-Pacific, equal country weights	$[\tau{+}1, \tau{+}21]$	-0.074^{***}	-0.036	-0.038^{**}	154,811
		(-5.85)		(-2.21)	
Asia-Pacific, equal country weights	months 2–3	-0.006	-0.004	-0.003	152,827
		(-0.50)		(-0.12)	
Asia-Pacific, equal country weights	months 4–6	-0.007	$+0.042$	-0.049^{***}	149,575
		(-0.53)		(-2.60)	

Notes: The within-cohort regression of Table IA.6, Panel A, applied to nineteen developed markets and the regional subsamples shown. Standard errors are two-way clustered by firm and outcome-ending calendar month. Rows labeled equal country weights reweight stock-months to give countries equal weight. β_1 is the coefficient on PreTOM return and β_2 the coefficient on Early return; -1 denotes full recovery of the observed relative window loss. Parentheses beneath β_1 and $\beta_1 - \beta_2$ report t-statistics. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Leave-one-country-out, pooled β_1 at $[\tau{+}1, \tau{+}3]$ / $[\tau{+}1, \tau{+}21]$: range -0.019 to -0.015 / -0.041 to -0.025.

Table IA.26: The PreTOM Window's Abnormal Share of the Momentum Premium across Levels, Samples, and Subperiods

	Full month	PreTOM (6 days)	Rest (15 days)	Raw share	Abnormal share, Rest
Panel A: All months, 1980–2025 (552 months)					
WML	96.7	60.9	35.8	63%	48%
Losers − Mkt	−60.9	−47.9	−13.0	79%	70%
Winners − Mkt	35.8	13.0	22.8	36%	11%
Shares, 95% bootstrap interval: raw [33%, 150%]; abnormal, Rest [7%, 170%]					
Panel B: Excluding the 2001–02, 2009, and 2020 crash episodes (531 months)					
	Full month	PreTOM (6 days)	Rest (15 days)	Raw share	Abnormal share, Rest
WML	116.1	69.2	46.9	60%	43%
Losers − Mkt	−78.0	−55.0	−22.9	71%	59%
Winners − Mkt	38.2	14.2	23.9	37%	12%
Shares, 95% bootstrap interval: raw [35%, 108%]; abnormal, Rest [9%, 111%]					

Table IA.26: The PreTOM Window's Abnormal Share of the Momentum Premium across Levels, Samples, and Subperiods (continued)

Panel C: 1927–2025 (1,188 months)					
	Full month	PreTOM (6 days)	Rest (15 days)	Raw share	Abnormal share, Rest
WML	123.8	49.5	74.3	40%	16%
Losers − Mkt	−68.3	−38.1	−30.1	56%	38%
Winners − Mkt	55.5	11.3	44.2	20%	−11%
Shares, 95% bootstrap interval: raw [26%, 57%]; abnormal, Rest [−4%, 40%]					
Panel D: Subperiods, WML in bps per month					
Sample	Full month	PreTOM (6 days)	Rest (15 days)	Raw share	
1980–2001 (264 months)	167.6***	61.5***	106.2***	37%	
	(4.15)	(3.29)	(2.99)		
1980–2001 ex crashes	168.3***	63.1***	105.2***	38%	
(259 months)	(4.25)	(3.37)	(3.01)		
2002–2025 (288 months)	31.7	60.4**	−28.7	n/a	
	(0.64)	(2.19)	(−0.66)		
2002–2025 ex crashes	66.6	75.1***	−8.5	n/a	
(272 months)	(1.56)	(2.95)	(−0.22)		

Notes: Value-weighted daily returns, fixed monthly sorting, in basis points. Full month is six times the mean PreTOM daily return plus fifteen times the mean daily return on the other days (Rest); Rest is the benchmark throughout this table. Raw share is PreTOM divided by Full month; the abnormal shares divide $6\times$(PreTOM rate$-$benchmark rate) by Full month. Intervals are percentile bootstraps over months. No share is reported for 2002–2025 because the full-month premium is not distinguishable from zero. In Panel D, t-statistics in parentheses are from monthly sums. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.22 Alternative Benchmarks

Section 3 of the paper attributes to the window its return in excess of an ordinary-day benchmark. This section reports the benchmarked excess under every ordinary-day benchmark we construct and under four accounting conventions, in excess terms out of sample, by era, and in the 19 developed markets. Table IA.27 reports the temporal benchmarks and the eras; Table IA.28 the international estimates; Table IA.11 the stock-month and date fixed-effect estimates.

Under the temporal benchmarks of Table IA.27 the WML excess is 46.6 basis points per month against Rest and 45.3 against Mid, the exact calendar interior of the month (every trading day after the previous Post window and before the current PreTOM window, whatever the month's length), 45.6 in a daily regression with month and day-of-week effects, and 49.5 with the crash episodes and the 0.5% tails of daily WML removed. The eight middle deciles earn no window excess over the interior (-0.7 basis points, $t = -0.3$), and using them rather than the market as the untreated group gives a loser contribution of 42.0 against Rest and 29.5 against Mid. Under the Rest benchmark the four accounting conventions give a WML excess of 46.6 (pooled daily means), 46.9 (equal-weighted monthly contrasts, time-series standard error 20.9), 46.9 (actual monthly day counts), and 51.7 (compounded returns), and shares of the full-month premium of 48%, 48%, 48%, and 52%. On 2001–2025, the Rest-benchmarked excess of every six-day block $[\tau-(s+5), \tau-s]$ is: $s = 0$, 46.9 (loser 15.3); $s = 1$, 51.4 (33.0); $s = 2$, 58.4 (31.4); $s = 3$, 73.5 (43.0); $s = 4$, the chosen window, 63.8 (46.1); $s = 5$, 40.9 (34.5); $s = 6$, 38.6 (43.1); $s = 7$, 21.2 (24.9); $s = 8$, 8.7 (14.7); $s = 9$, -19.2 (-1.8). The institutional-era minus pre-institutional difference in the loser contribution (1980–2025 minus 1927–1961) is 41.8 basis points per month with a standard error of 21.9; for WML it is 58.9 (29.4).

Table IA.27: The Window's Benchmarked Contribution Under Alternative Benchmarks

Panel A: Temporal benchmarks, 1980–2025 (bps per month)						
	WML excess		Loser contribution		Winner contribution	
Benchmark	Estimate	t	Estimate	t	Estimate	t
Rest (all other days)	$+46.6^{**}$	2.25	$+42.7^{***}$	2.73	$+3.9$	0.39
Mid (day 4 to $\tau-10$)	$+45.3^{**}$	2.13	$+30.3^{*}$	1.86	$+15.0$	1.60
Weekday-adjusted (month + day-of-week FE)	$+45.6^{**}$	2.17	$+41.4^{***}$	2.59	$+4.2$	0.44
Month FE only	$+46.5^{**}$	2.23	$+42.6^{***}$	2.69	$+3.9$	0.40
Outlier-clean (no crash episodes, 1% WML tails)	$+49.5^{***}$	2.83	$+44.4^{***}$	3.50	$+5.1$	0.58
Outlier-clean, Mid benchmark	$+45.8^{**}$	2.47	$+28.8^{**}$	2.14	$+16.9^{*}$	1.81
Panel B: Eras, Rest benchmark						
1927–1961	-12.6	-0.58	$+1.8$	0.09	-14.4	-1.27
1962–1979	$+16.9$	0.87	$+32.7^{***}$	2.78	-15.8	-1.35
1980–2025	$+46.6^{**}$	2.25	$+42.7^{***}$	2.73	$+3.9$	0.39

Notes: Value-weighted portfolios with fixed monthly sorting. Contributions equal six times the PreTOM−benchmark daily return difference, in basis points per month, with the loser sign reversed for its WML contribution. Displayed contributions use pooled daily means; t-statistics use the corresponding paired monthly contrasts. Rest comprises all non-PreTOM days; Mid runs from trading day four through $\tau-10$. Regression-adjusted rows use the fixed effects stated. Outlier-clean rows exclude the 2001–02, 2009 and 2020 episodes and the top and bottom 0.5% of daily WML returns (1% in total). Panel B uses Rest by era. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.28: The Benchmarked Window Excess in 19 Developed Markets

	Benchmark: all other days (Rest)			Benchmark: mid-month days (Mid)		
	WML excess	Loser contribution	Winner contribution	WML excess	Loser contribution	Winner contribution
Pooled, inverse-variance weights	+32.5 (14.0)	+19.4 (8.2)	+12.5 (7.9)	+35.9 (15.3)	+18.1 (9.1)	+17.1 (8.7)
Pooled, equal country weights	+31.8 (7.3)	+17.3 (6.4)	+14.5 (5.7)	+25.2 (10.9)	+15.0 (7.2)	+10.2 (7.1)
Leave-one-country-out range	[+29.3, +35.1]	[+17.5, +23.5]	[+10.9, +13.8]	[+33.3, +39.8]	[+15.9, +22.0]	[+15.3, +19.0]
Countries with the predicted sign	16/19	14/19	15/19	14/19	13/19	14/19
Japan (reference, not pooled)	−21.8 (21.8)	−0.8 (13.1)	−20.9 (12.0)	+6.6 (24.0)	+13.5 (14.0)	−6.9 (13.7)

Table IA.28: The Benchmarked Window Excess in 19 Developed Markets (continued)

	Benchmark: all other days (Rest)			Benchmark: mid-month days (Mid)		
	WML excess	Loser contribution	Winner contribution	WML excess	Loser contribution	Winner contribution
NOR	+65.5	+61.8	+3.7	+76.5	+68.8	+7.8
ESP	+88.0	+61.0	+27.1	+88.5	+53.9	+34.6
FIN	+61.2	+43.6	+17.6	+68.5	+46.7	+21.8
CHE	+66.6	+35.5	+31.2	+57.1	+29.9	+27.3
AUS	+31.5	+33.8	−2.3	+48.5	+39.9	+8.6
HKG	+22.6	+32.4	−9.8	+29.5	+41.9	−12.5
GBR	+38.7	+32.3	+6.4	+37.2	+20.7	+16.5
FRA	+49.9	+31.0	+18.9	+35.1	+19.8	+15.2
BEL	+13.0	+28.5	−15.5	+19.4	+25.7	−6.3
DNK	+35.9	+25.1	+10.8	+40.4	+22.5	+17.9
DEU	+31.5	+15.3	+16.2	+28.0	+10.1	+18.0
NLD	+26.9	+15.0	+12.0	+42.3	+25.6	+16.7
SWE	+46.0	+12.7	+33.4	+75.4	+25.1	+50.3
ITA	+1.3	+1.0	+0.3	−6.8	−5.9	−0.9
AUT	−17.6	−0.1	−17.6	−102.4	−9.4	−93.0
SGP	+11.6	−15.7	+27.3	+25.3	−18.9	+44.2
NZL	−16.7	−18.2	+1.5	−59.5	−38.1	−21.4
IRL	+71.1	−26.7	+97.8	−7.4	−37.4	+30.0
PRT	−22.6	−39.8	+17.2	−16.7	−36.3	+19.6

Notes: Value-weighted decile portfolios by country from Compustat Global, 1991–February 2026, in excess of each country's own value-weighted market, in basis points per month. Each entry is $6 \times$ (PreTOM rate − benchmark rate) for WML and for each leg, with the sign of the leg's contribution to WML, computed as a paired monthly contrast within country and averaged over months; standard errors in parentheses are time-series standard errors of the monthly contrasts. Standard errors are clustered by month for the inverse-variance row; the equal-country row treats country estimates as independent. All three series are estimated on the country-days where the loser leg, the winner leg and WML are all observed, so that the WML excess equals the sum of the two signed leg contributions within each country before pooling. Rest is every non-PreTOM trading day of the country's own calendar; Mid is the exact interior of the calendar month, the days after the previous Post window and before the current PreTOM window, so its length varies with the month. Japan is reported separately and excluded from the pooled estimates.

Market beta. Table IA.29 reports the CAPM-adjusted loser contribution of Table IA.12 under alternative estimates of the loser beta. Excluding PreTOM days leaves the full-sample beta essentially unchanged (1.33 against 1.31), so the window itself does not inflate it. Betas estimated from trailing data are lower, 1.17 to 1.19, and leave a larger contribution against Rest, 29.9 to 34.2 basis points per month, significant at the 5 percent level. The main text uses the trailing one-year beta. The two full-sample betas give smaller contributions against Rest, +23.5 and +24.3 basis points per month, neither statistically significant ($t = 1.6$). Against Mid, the contribution is not significant under any estimate.

Table IA.29: CAPM-Adjusted Loser Contribution Under Alternative Beta Estimates

		Loser contribution, CAPM-adjusted	
Loser beta estimated from	Mean β	Rest	Mid
Full sample	1.31	+24.3	+16.8
		(1.6)	(1.1)
Full sample,	1.33	+23.5	+16.2
excluding PreTOM days		(1.6)	(1.1)
Trailing one year	1.17	+30.2**	+15.7
(Table IA.12)		(2.1)	(1.1)
Trailing two years	1.19	+29.9**	+15.8
		(2.0)	(1.1)
Trailing two years, skipping	1.19	+34.2**	+21.6
the most recent year		(2.1)	(1.3)

Notes: The CAPM-adjusted loser contribution of Table IA.12, column 5, under five estimates of the loser leg's market beta: the full 1980–2025 sample; the full sample excluding PreTOM days; trailing windows of 252 and 504 trading days ending the day before each return; and a 504-day window ending 252 trading days before each return. Trailing betas use only information available before the day they adjust. Mean β is the time-series average. Contributions are six times the PreTOM-minus-benchmark daily difference of the loser return less β times the market return, in basis points per month, with the sign of its contribution to WML; value-weighted, fixed monthly sorting, 1980–2025. t-statistics in parentheses are from paired monthly contrasts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.23 Acute Liquidity Shocks

This section describes how the two momentum legs behaved in three selected episodes of acute aggregate liquidity demand in 1998–2025: September 2008, October 2008 and March 2020. Everything measured here is monthly, so the episodes say nothing about where within the month the asymmetry falls and cannot bear on whether it arises inside or outside the window. These monthly comparisons do not locate the selling within the month. Table IA.30 reports the episodes. In each, momentum losers fell heavily relative to winners (Panel A). At the fund level, sorting actively managed U.S. equity funds each month-end into deciles by trailing twelve-month return, past-loser funds underperformed their Carhart four-factor benchmark by 11.17 percentage points in March 2020 while past-winner funds were at their benchmark (-0.26), with industry outflows of 1.49% of assets; the 2008 episodes show smaller abnormal spreads but economically large abnormal returns for loser funds (-3.22 and -3.90 percentage points; Panel B). The fund-level asymmetry runs in the same direction as the stock-level one.

Table IA.30: Liquidity Shocks and the Loser Asymmetry

Panel A: Stock-Level Momentum Portfolio Returns During Liquidity Crisis Episodes

Episode	(1) D1 (Losers) (%)	(2) D10 (Winners) (%)	(3) D10−D1
Sep 2008 (GFC)	−25.90	−15.42	+10.48
Oct 2008 (GFC)	−26.05	−15.12	+10.93
Mar 2020 (COVID)	−27.96	−9.77	+18.19

Panel B: Momentum Fund Returns During the Same Three Episodes

Episode	(1) D1 excess (%)	(2) D1 Carhart abnormal (%)	(3) D10 excess (%)	(4) D10 Carhart abnormal (%)	(5) D10−D1 Carhart abnormal
Sep 2008 (GFC)	−11.56	−3.22	−12.52	−1.99	1.23
Oct 2008 (GFC)	−21.89	−3.90	−16.01	−1.36	2.54
Mar 2020 (COVID)	−24.92	−11.17	−11.29	−0.26	10.92

Notes: Panel A reports compounded value-weighted monthly returns on stock-level momentum decile portfolios (NYSE breakpoints, fixed monthly sorts on cumulative returns over months −12 through −2) during three acute-outflow episodes; the D10−D1 column is the percentage-point spread between winner and loser deciles. Panel B reports value-weighted (lagged TNA) monthly returns on D1 and D10 mutual fund portfolios during the same three episodes. Aggregate mutual fund net outflows during these episodes were 0.90% (September 2008), 1.79% (October 2008), and 1.49% (March 2020) of one-month-lagged assets. Funds are sorted at the end of each month into deciles on their trailing 12-month return (months $[m-12, m-1]$, no skip); the decile assignment applies to month m's return. The "excess" columns report monthly returns in excess of the one-month T-bill rate; the "Carhart abnormal" columns report the residual after subtracting the return implied by a Carhart (1997) four-factor model (MKT, SMB, HML, UMD), with loadings estimated on the 322 non-crisis months in the 1998–2025 sample. The D10−D1 Carhart abnormal column is the difference of the two abnormal-return columns, in percentage points. Sample for Panel B: CRSP Mutual Fund Database, actively managed U.S. equity diversified funds (objective code ED, AUM $\geq$ \$10M, excluding index funds, ETFs, and ETNs) that appear in either D1 or D10 in at least one month between 1998 and 2025; all months for those funds are retained.

IA.24 The Momentum Factor and Fund Persistence

Carhart (1997) attributes most mutual-fund performance persistence to momentum exposure and to persistent differences in investment costs. One residual survives that accounting, and it sits at the bottom of the distribution: the worst-return funds keep underperforming by more than loadings and fees explain. This section is about the two sources he does identify, and specifically about when the first of them is realized.

A negative momentum loading can come from holding losers or from avoiding winners, so we measure exposure to loser stocks directly. If the momentum factor's premium is concentrated in PreTOM, negative exposure has a corresponding concentration in its return contribution, while fees accrue throughout the month. These different calendar patterns provide an additional way to distinguish the components. The evidence here comes from two sources the rest of the paper does not use, Kenneth French's published daily momentum factor and daily net returns from the CRSP Mutual Fund Database.

IA.24.1 The Momentum Factor's Intramonth Profile

The published momentum factor earns 5.51 basis points per day in PreTOM ($t = 3.58$) against 1.36 on the other fifteen trading days ($t = 1.24$), 1980 to 2025. A dollar invested in the factor on window days alone grows to \$5.51 over the sample, against \$2.26 on the other days and \$12.44 on all of them (Table IA.31, Panel A).

The factor is Kenneth French's own series, rebalanced daily rather than held fixed within the month and built on his universe and his breakpoints. Its window concentration therefore does not follow from the construction of Section 2. It also fixes the calendar on which a negative momentum loading costs a fund, since any fund carrying one bears most of the corresponding drag on the same six days.

IA.24.2 Loser-Fund Underperformance within the Month

We estimate the fund regression at daily frequency, adding the loser-stock return as a fourth regressor. For each fund-performance decile Dk, we estimate the regression separately on

PreTOM days, Rest days and the full sample. Index these estimation samples by w:

$$r_d^{Dk} - r_d^f = \alpha_w^{Dk} + \beta_w^{Dk}\left(r_d^{mkt} - r_d^f\right) + s_w^{Dk}\mathrm{SMB}_d + h_w^{Dk}\mathrm{HML}_d + \gamma_w^{Dk} r_d^{\mathrm{Loser}} + \varepsilon_{d,w}^{Dk}, \qquad d \in w, \text{ (IA.8)}$$

where r_d^{Dk} is the equal-weighted day-d net return of funds in decile Dk, after fees and expenses, computed from daily net asset values and distributions in the CRSP Mutual Fund Database; r_d^f is the daily one-month Treasury-bill rate; and r_d^{Loser} is the daily value-weighted excess return on the bottom stock-momentum decile. Setting $\gamma_w^{Dk} = 0$ gives the corresponding three-factor specification on that window. All slopes, as well as the intercept, are specific to the estimation window w. Letting it differ from zero asks whether exposure to losing stocks accounts for that alpha.[6]

In the window, past-loser funds earn a three-factor alpha of -2.02 basis points per day ($t = -1.57$), which falls to -0.38 ($t = -0.36$) once the loser-stock return enters. On the other days the same alpha moves only from -2.40 ($t = -2.65$) to -1.82 ($t = -2.47$). Loser-stock exposure removes 81 percent of the window alpha and 24 percent of the alpha on the other days (Table IA.31, Panel B).

The winner side moves the same way. Past-loser funds load on the loser-stock portfolio at $+0.28$ and past-winner funds at -0.19, and the winner funds' positive window alpha also disappears under the control. One channel therefore runs through both ends of the fund sort, since loser-stock returns cost the funds that hold those stocks and spare the funds that avoid them.

IA.24.3 Expense Ratios and the Fund-Day Panel

Expense ratios differ across funds, so separating them from the momentum component calls for a panel in which each fund is its own observation rather than a portfolio time series. Table IA.31, Panel C, relates past-loser fund returns to loser-stock exposure and expense ratios on fund-days. With both terms present, past-loser funds have no detectable alpha in either part of the month, $+1.85$ basis points per day in the window ($t = 1.23$) and -1.10 outside it ($t = -1.11$). The estimated intercepts are not distinguishable from zero once

[6] On the original 1963–1993 sample our monthly replication of Carhart matches his. Extended through 2025 the D10 minus D1 spread attenuates, as Choi and Zhao (2021) document. Daily fund returns implied by net asset values begin in September 1998.

both terms are included. Because expense ratios enter uncentered, these intercepts refer to a zero-fee fund.

IA.24.4 The Settlement Reform in the Factor and in Fund Returns

The settlement reform of Section 4 tests the same mapping a third time. We re-estimate equation (4) with the daily momentum factor, and then with the equal-weighted past-loser fund portfolio, on the left-hand side, keeping the $[\tau-4, \tau-3]$ treated days and the $[\tau'+5, \tau'+8]$ control, trading days 5 to 8 of the month. The factor's boundary-day difference-in-differences is -78.1 basis points ($t = -2.48$), negative because the factor is long winners and short losers, and the fund portfolio's is $+29.9$ ($t = 2.28$) with three-factor controls and $+51.6$ ($t = 2.03$) without (Table IA.32).

Alternative inference. For the monthly boundary-spread regressions, Newey–West standard errors with three lags give t-statistics of -2.36 for UMD and 2.93 for the uncontrolled loser-fund portfolio. Clustering the daily FF3-controlled fund regression by calendar month, so that the two boundary days of a month are not treated as independent, gives a standard error of 14.26 and $t = 2.10$.

The regulatory change that moves stock-level loser underperformance one day closer to month-end moves the published momentum factor and past-loser fund returns by one day as well. Carhart (1997) attributes fund-performance persistence to momentum exposure and expenses, leaving the worst funds' persistent underperformance as the residual his four-factor model does not reach. The evidence here adds when the momentum-related component is realized, and what moves it.

Table IA.31: UMD Factor and Carhart Decomposition at Daily Frequency

Panel A: Daily UMD returns by calendar window, 1980–2025

	Mean (bps/day)	t-stat	\$1 cumulative	Days
PreTOM ($[\tau-9, \tau-4]$)	5.51	3.58	\$5.51	3,312
Rest of month	1.36	1.24	\$2.26	8,283
Full month	2.55	2.84	\$12.44	11,595

Panel B: D1 and D10 fund daily alphas (bps/day) by calendar window, with loser-stock excess return control (1998–2025)

	D1 (Losers)		D10 (Winners)	
Specification	α	β_{Loser}	α	β_{Loser}
PreTOM only				
FF3 only	-2.02		1.49	
	(-1.57)		(1.30)	
FF3 + Loser	-0.38	0.28^{***}	0.39	-0.19^{***}
	(-0.36)	(10.88)	(0.37)	(-9.38)
Observations	1,968	1,968	1,968	1,968
Rest of the month				
FF3 only	-2.40^{***}		1.31^{*}	
	(-2.65)		(1.69)	
FF3 + Loser	-1.82^{**}	0.28^{***}	0.93	-0.18^{***}
	(-2.47)	(18.01)	(1.36)	(-13.06)
Observations	4,908	4,908	4,908	4,908
Full sample				
FF3 only	-2.28^{***}		1.32^{**}	
	(-3.10)		(2.06)	
FF3 + Loser	-1.41^{**}	0.28^{***}	0.75	-0.18^{***}
	(-2.32)	(20.36)	(1.32)	(-15.49)
Observations	6,876	6,876	6,876	6,876

Notes: Panel A decomposes the Fama–French daily UMD factor into PreTOM and Rest-of-month components, 1980–2025; t-statistics use Newey–West HAC standard errors with automatic bandwidth. Panel B reports estimates of equation (IA.8) for funds in the loser (D1) or winner (D10) decile, 1998–2025. The LHS is the equal-weighted daily fund excess return (fund return minus the one-month T-bill rate) in basis points; daily fund returns outside ±50% are dropped. The loser-stock control is the value-weighted excess return on the bottom CRSP momentum decile of stocks (also in fund-return form: stock return minus r^f), entered as a level—not residualized against the FF3 factors. HAC(5) Newey–West time-series regressions; β_{Loser} measures the fund portfolio's loading on the loser-stock excess return. t-statistics are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table IA.31: UMD Factor and Carhart Decomposition at Daily Frequency (continued)

Panel C: D1 fund alpha, stepwise absorption by the loser return and the expense ratio			
	(1) FF3	(2) +Loser	(3) +Loser+Exp
PreTOM days (N = 1,125,059)			
α (bps/day)	-1.87	-0.19	$+1.85$
	(-1.56)	(-0.19)	(1.23)
β_{Loser}	—	$+0.264^{***}$	$+0.270^{***}$
	—	(15.18)	(14.95)
Exp β (per 1%)	—	—	-1.48^{*}
	—	—	(-1.95)
Rest of the days (N = 2,806,032)			
α (bps/day)	-2.35^{***}	-1.95^{***}	-1.10
	(-2.98)	(-3.00)	(-1.11)
β_{Loser}	—	$+0.259^{***}$	$+0.267^{***}$
	—	(26.05)	(25.76)
Exp β (per 1%)	—	—	-0.59
	—	—	(-1.19)

Notes (Panel C): D1 fund-day regressions, 1998–2025, using Panel B's fund universe and return filter. Expense ratios are annual percentage points and enter uncentered; intercepts therefore refer to a zero expense ratio. Parentheses report t-statistics with standard errors clustered by date. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table IA.32 reports the settlement estimates of Section IA.24.4. Panel A reports the difference-in-differences on the daily momentum factor and Panel B the same specification on the equal-weighted past-loser fund portfolio.

Table IA.32: Settlement Change Difference-in-Differences for UMD and Loser Funds

Panel A: T+1 settlement DiD on the Fama–French daily UMD factor (1980–2025)	Coefficient
Intercept ($[\tau'+5, \tau'+8]$ base)	2.39
	(1.18)
$\mathbf{1}_{[\tau-4]}$	2.87
	(0.75)
$\mathbf{1}_{[\tau-3]}$	0.42
	(0.10)
T1Reform_t	−1.62
	(−0.13)
$\theta_{\tau-4}$ ($\text{T1Reform}_t \times \mathbf{1}_{[\tau-4]}$)	−42.63
	(−1.61)
$\theta_{\tau-3}$ ($\text{T1Reform}_t \times \mathbf{1}_{[\tau-3]}$)	35.45
	(1.46)
DiD ($\theta_{\tau-4} - \theta_{\tau-3}$)	−78.08**
	(−2.48)
Monthly boundary spread, difference in means	−78.08**
	(−2.00)
Observations	3,303

Notes: Boundary-day DiD on the Fama–French daily UMD factor (in basis points), parallel to the fund Panel B specification. Sample is restricted to treated days $\tau-4$ and $\tau-3$ and controls $[\tau'+5, \tau'+8]$, trading days 5–8 of the month, 1980–2025, excluding May 2024 to avoid contamination at the regime boundary. T1Reform_t equals one for dates from June 1, 2024 onward, after the T+1 settlement reform took effect. Parentheses on daily estimates report heteroskedasticity-robust t-statistics. The contrast is $\theta_{\tau-4} - \theta_{\tau-3}$; it is negative because UMD is long winners and short losers, so selling pressure leaving the old boundary day raises the factor there. This mirrors the stock-level shift documented in Table 8 and the fund-level shift documented in Panel B below. The final row regresses the monthly $\tau-4$ minus $\tau-3$ spread on the reform indicator, using 532 pre-reform and 19 post-reform months with heteroskedasticity-robust (HC2) inference. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table IA.32: Settlement Change Difference-in-Differences for UMD and Loser Funds (continued)

Panel B: T+1 settlement DiD on D1 fund portfolio (CRSP daily, 1998–2025)		
	(1) Simple	(2) With FF3
Intercept	−3.70	−2.50
	(−0.96)	(−1.43)
T1Reform_t	19.00	−1.25
	(1.15)	(−0.26)
$\mathbf{1}_{[\tau-4]}$	0.09	0.23
	(0.01)	(0.06)
$\theta_{\tau-4}$ ($\text{T1Reform}_t \times \mathbf{1}_{[\tau-4]}$)	12.22	6.27
	(0.49)	(0.57)
$\mathbf{1}_{[\tau-3]}$	19.80**	0.33
	(2.36)	(0.10)
$\theta_{\tau-3}$ ($\text{T1Reform}_t \times \mathbf{1}_{[\tau-3]}$)	−39.36	−23.64**
	(−1.64)	(−2.40)
DiD ($\theta_{\tau-4} - \theta_{\tau-3}$)	51.58**	29.91**
	(2.03)	(2.28)
Monthly boundary spread	51.58**	
	(1.97)	
Observations	1,959	1,959

Notes: Equation (4) applied to equal-weighted D1 fund returns, 1998–2025. Treated days are $\tau-4$ and $\tau-3$; controls are $[\tau'+5, \tau'+8]$, trading days 5–8 of the month. May 2024 is excluded and the post-period begins in June. Column (2) adds FF3 controls. Parentheses on daily estimates report t-statistics with standard errors clustered by date. The final row uses the uncontrolled monthly boundary spread, with 308 pre-reform and 19 post-reform months and heteroskedasticity-robust (HC2) inference; it corresponds to column (1). Alternative inference is reported in Section IA.24.4. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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